Microsoft Certified Azure Administrator: AZ-104

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TECHNOLOGY

Configure and Manage Virtual Networking

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A Day in the Life of an Azure Architect

Amidst company migration to Azure, an Azure Administrator wants to replicate the on-premise network to the cloud. The Azure resources must be organized into virtual networks and subnets. The company requires an Azure IP addressing schema. The schema should provide flexibility, room for growth, and integration with on-premises networks to enable a complete networking setup.

The Azure Administrator needs to design the apps to be resilient to failure and scale easily when the demands increase.

Resiliency is to be improved by adding multiple web servers to its site and distributing the load across them. Centralize the site on a single load-balancing service. This will simplify the URLs for site visitors.

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Learning Objectives

By the end of this lesson, you will be able to:

- Assess the importance and methodology of virtual network planning, subnetting, and IP addressing to optimize Azure infrastructure
- Analyze network connectivity issues to determine root causes and implement effective solutions in Azure
- Show proficiency in deploying and utilizing Azure Bastion to enhance secure remote access
- Describe the purpose and basic operations of Network Security Groups (NSG) to maintain secure Azure environments

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Administer Virtual Networking

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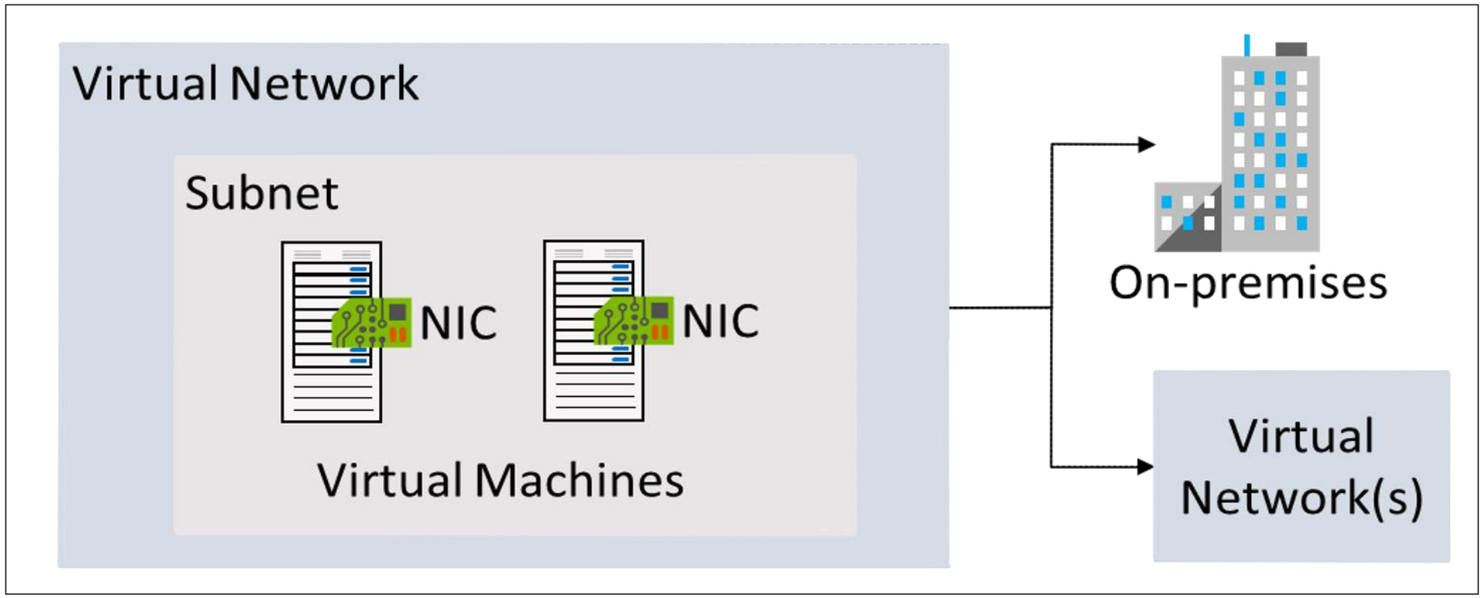
Virtual Networks

Features:

- Represents the network logically
- Creates a dedicated private cloud-only virtual network
- Extends datacenter with virtual networks securely
- Enables hybrid cloud scenarios



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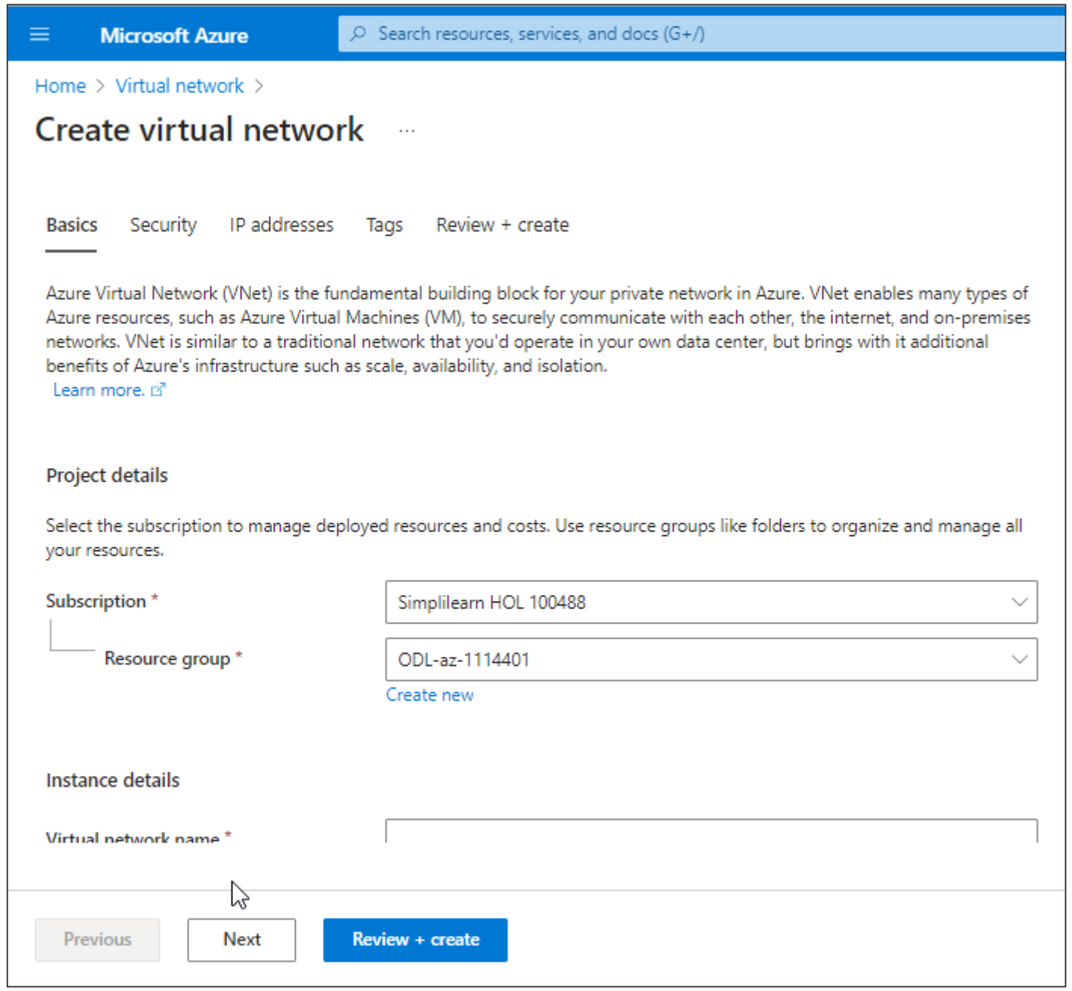
Create Virtual Networks



Tips:

- Create new virtual networks any time
- Add virtual networks when creating a virtual machine
- Define the address space and at least one subnet
- Look out for overlapping address spaces

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Create Subnets

Consider the following points:

- A virtual network can be segmented into one or more subnets.
- Subnets provide logical divisions within the network. It increases security and performance and simplifies network management.
- Each subnet needs a unique address and must not overlap with the VNet subscription.

+ Subnet | + Gateway subnet | Refresh | Manage users | Delete

Name	IPv4	IPv6	Available IPs	Delegated
subnet0	10.0.0.0/24	-	250	-
subnet1	10.0.1.0/24	-	251	-
subnet2	10.0.2.0/24	-	251	-
AzureBastionSubnet	10.0.30.0/27	-	27	-
GatewaySubnet	10.0.3.0/27	-	availability dependent on dynamic use	-

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Creating Subnet and VNet

Duration: 10 Min.

Problem Statement:

You have been tasked with creating subnets inside a virtual network in the portal.

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Assisted Practice: Guidelines

Steps to be followed:

1. Create a virtual network in the portal

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Implement Azure Bastion

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Azure Bastion

It is a service that, when deployed, allows users to access a virtual machine through your browser and the Azure portal or using the preexisting SSH or RDP client on the local machine.

- It is a fully managed PaaS service that is deployed within the virtual network.
- It provides secure and seamless RDP/SSH connectivity to the VM directly through TLS.
- Once the connection is established, the VM doesn't require a public IP address, agent, or specific client software.
- It offers secure RDP/SSH connectivity to all VMs.

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Azure Bastion: Key Benefits

1. RDP/SSH session can be accessed with a click.
2. The RDP/SSH session is over TLS, and this enables the traffic to pass through firewalls more securely.
3. Azure Bastion establishes the RDP/SSH connection to the Azure VM using its private IP address. There's no need for a public IP address on the VM.
4. Azure Bastion connects to the VM using the private IP address, so NSG can be configured to allow RDP/SSH from Azure bastion.

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Azure Bastion: Key Benefits

5 It is a fully managed PaaS service from Azure with internal hardening to ensure secure RDP/SSH connectivity.

6 Since there is no need to expose your VMs to the internet, they are shielded from port scanning by malicious users.

7 It is positioned at the virtual network's perimeter, eliminating the need to harden each VM within that network.

8 It connects to the VM using the private IP address, so NSG can be configured to allow RDP/SSH from Azure bastion.

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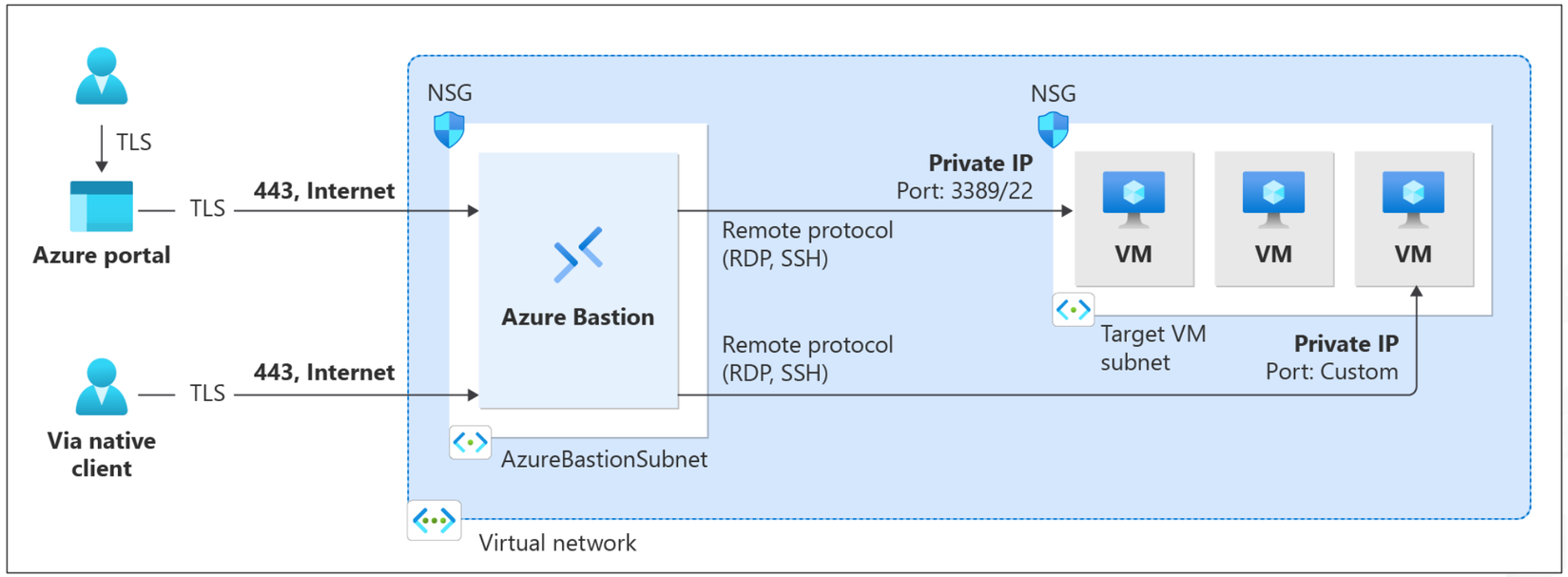
Azure Bastion: Architecture



- Bastion is deployed in the virtual network which includes the AzureBastionSubnet subnet, having at least a /26 prefix.
- Any HTML5 browser can be used to access the Azure portal.
- VM for connection needs to be chosen.
- RDP/SSH session opens in the browser with a single click.
- The Azure VM doesn't need a public IP.

Source: https://learn.microsoft.com/en-us/azure/bastion/bastion-overview

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Deploy Azure Bastion with Default Settings

Step 01

Deploy Bastion using default settings from the VM resource via the Azure portal

Step 02

Connect to VM through the portal using RDP/SSH and the VM's private IP address

Step 03

In case the VM has a public IP address that is not required, then remove it

Once Bastion is deployed, connect (SSH/RDP) to the VM within the virtual network through Bastion using the VM's private IP address.

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IP Addressing

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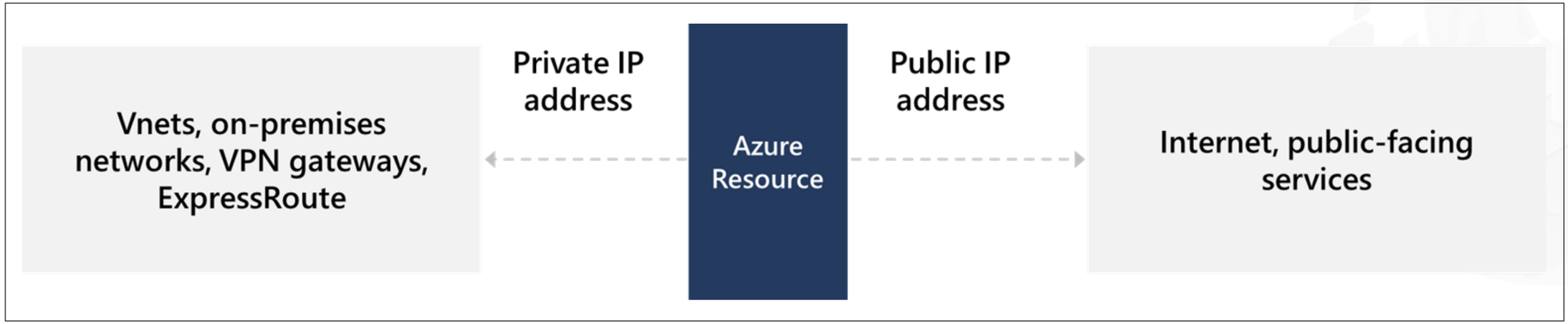
Plan IP Addressing

Private IP addresses:
It is used within an Azure virtual network (VNet) and on-premises network when using a VPN gateway or ExpressRoute circuit to extend the network to Azure.

Public IP addresses:
It is used for communication with the internet, including Azure public-facing services.



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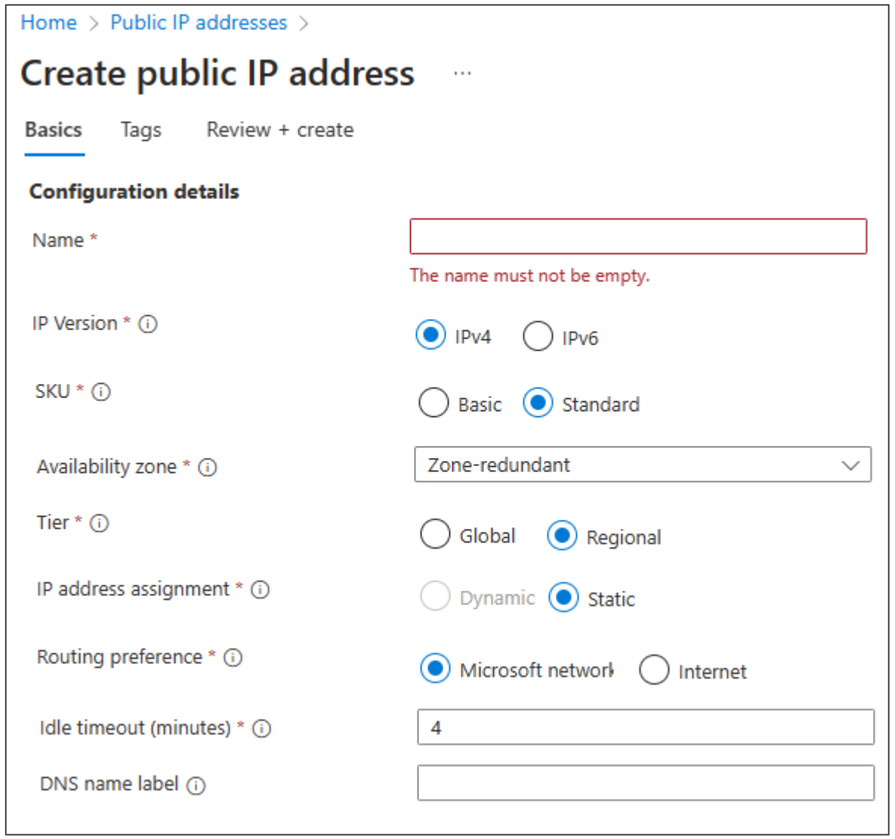
Create Public IP Addresses

Public IP is used to communicate with internet devices, including Azure public-facing services.



- Available in IPv4 or IPv6 or both
- Basic vs. Standard SKU
- Dynamic vs. Static
- Zone redundant (Standard SKU)
- Range of contiguous addresses available as a prefix

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Associate Public IP Addresses

A public IP address resource can be associated with virtual machine network interfaces, internet-facing load balancers, VPN gateways, and application gateways.

Top-level resource	IP address association	Dynamic IPV4	Static IPV4	Dynamic IPV6	Static IPV6
Virtual machine	Network interface	Yes	Yes	Yes	Yes
Public load balancer	Front-end configuration	Yes	Yes	Yes	Yes
VPN gateway	Gateway IP configuration	Yes (non-AZ only)	Yes	No	No
Application gateway	Front-end configuration	Yes (V1 only)	Yes (V2 only)	No	No

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Associate Private IP Addresses

- Dynamic (default): Azure assigns the next available unassigned or unreserved IP address in the subnet's address range.
- Static: Azure selects and assigns any unassigned or unreserved IP address in the subnet's address range.

Private IP addresses	IP address association	Dynamic	Static
Virtual Machine	NIC	Yes	Yes
Internal Load Balancer	Front-end configuration	Yes	Yes
Application Gateway	Front-end configuration	Yes	Yes

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Network Security Group (NSG)

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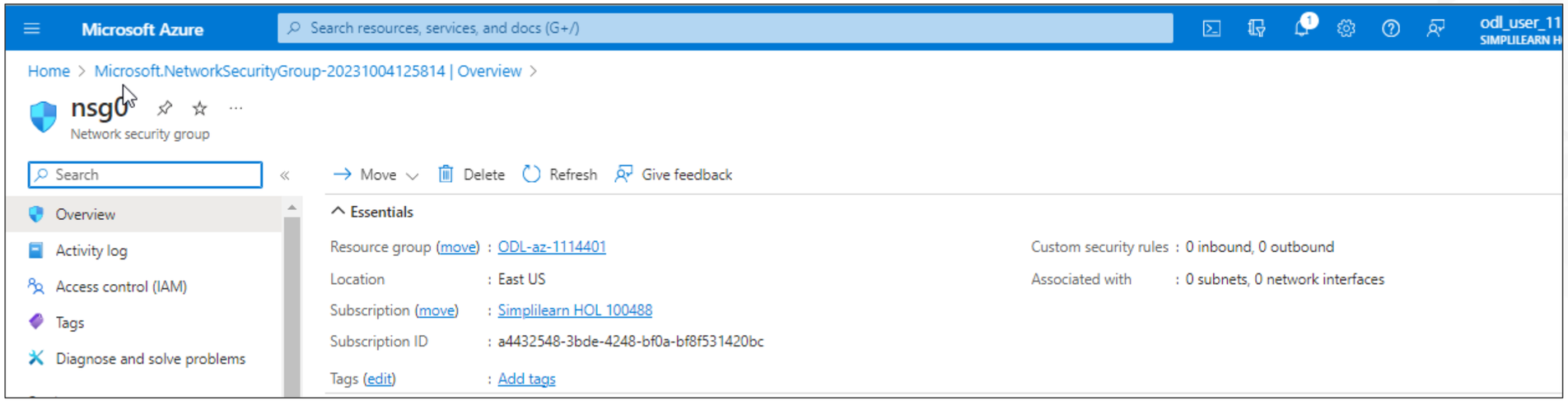
Implement Network Security Groups

Points to consider while implementing NSG:

- Limits network traffic to resources in a virtual network
- Lists the security rules that allow or deny inbound or outbound network traffic
- Associates multiple times to a subnet or a network interface



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Determine NSG Rules

- Security rules in NSGs enable filtering network traffic that can flow in and out of virtual network subnets and network interfaces.
- There are default security rules, but other rules with higher priority can be added.

Inbound security rules

Priority	Name	Port	Protocol	Source	Destination	Action
100	RDP_Inbound	3389	Any	Any	Any	Allow
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowAzureLoadBalancerInBound	Any	Any	AzureLoadBalancer	Any	Allow
65500	DenyAllInBound	Any	Any	Any	Any	Deny

Outbound security rules

Priority	Name	Port	Protocol	Source	Destination	Action
65000	AllowVnetOutBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowInternetOutBound	Any	Any	Any	Internet	Allow
65500	DenyAllOutBound	Any	Any	Any	Any	Deny

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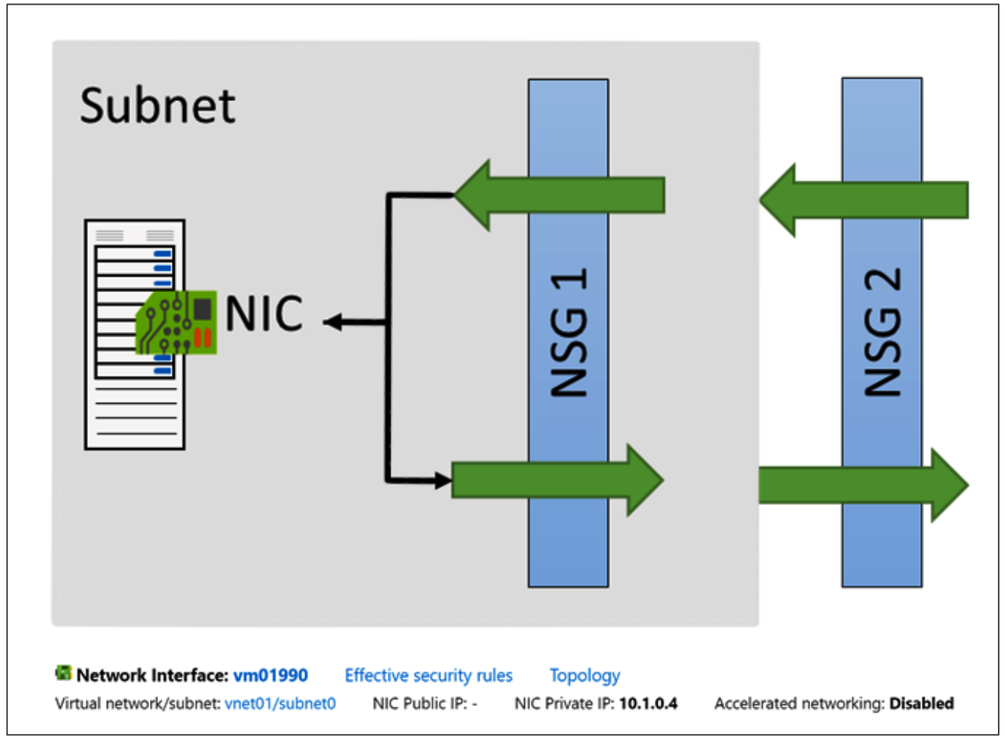
Determine NSG Effective Rules

- NSGs are evaluated independently for the subnet and NIC.
- An allow rule must exist at both levels for traffic to be admitted.
- The Effective Rules link can be used if there is any confusion regarding the applied security rules.



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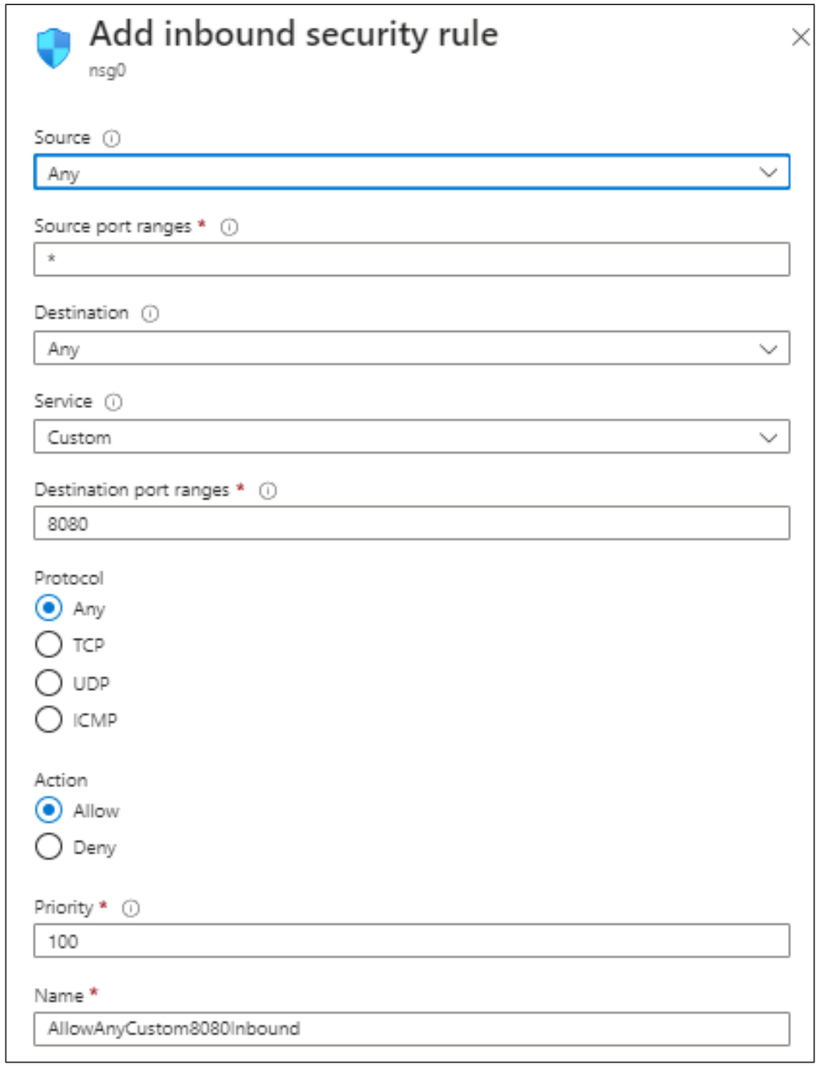
Create NSG rules



To feature inbound and outbound rules, users will be able to select from styles of services mentioned below:

- Source or Destination: Any or an individual IP address, classless inter-domain routing (CIDR), service tag, or application security group
- Service: Custom, HTTPS, SSH, RDP, and DNS
- Priority: The lower the number, the higher the priority

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Creating Network Security Group

Duration: 10 Min.

Problem Statement:

You've been tasked with creating a network security group for controlling inbound and outbound traffic.

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Assisted Practice: Guidelines

Steps to be followed:

1. Create a network security group in the portal
2. Create inbound rules

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Application Security Group (ASG)

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Application Security Group

- It allows integration of network security seamlessly into an application's framework.
- This allows the grouping of VMs and establishes network security policies based on these groups.
- The user can focus on the business logic and the platform handles the complexity of explicit IP addresses and multiple rule sets.

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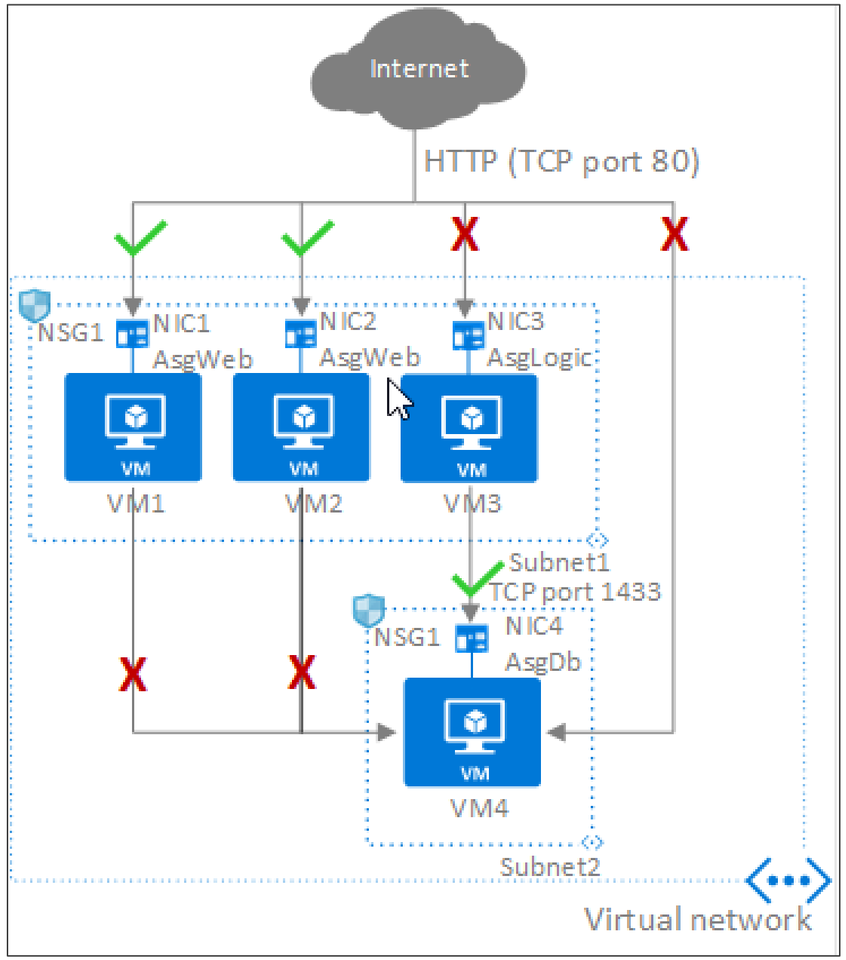
Application Security Group: Example



NSG1 is associated to both subnets and holds the following rules:

- Allow-HTTP-Inbound-Internet
- Deny-Database – All
- Allow-Database-BusinessLogic

Source: https://learn.microsoft.com/en-us/azure/virtual-network/application-security-groups

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Identity Domains and Custom Domains

Azure Domain Name Service (DNS) enables hosting DNS records for domains on Azure infrastructure. With Azure DNS, the same credentials, APIs, tools, and billing can be used as other Azure services.

- Creating an Azure subscription creates a Microsoft Entra ID domain for the user.
- The domain has an initial domain name in the form domainname.onmicrosoft.com.
- The user can customize or change the name.
- After the custom name is added, it must be verified.

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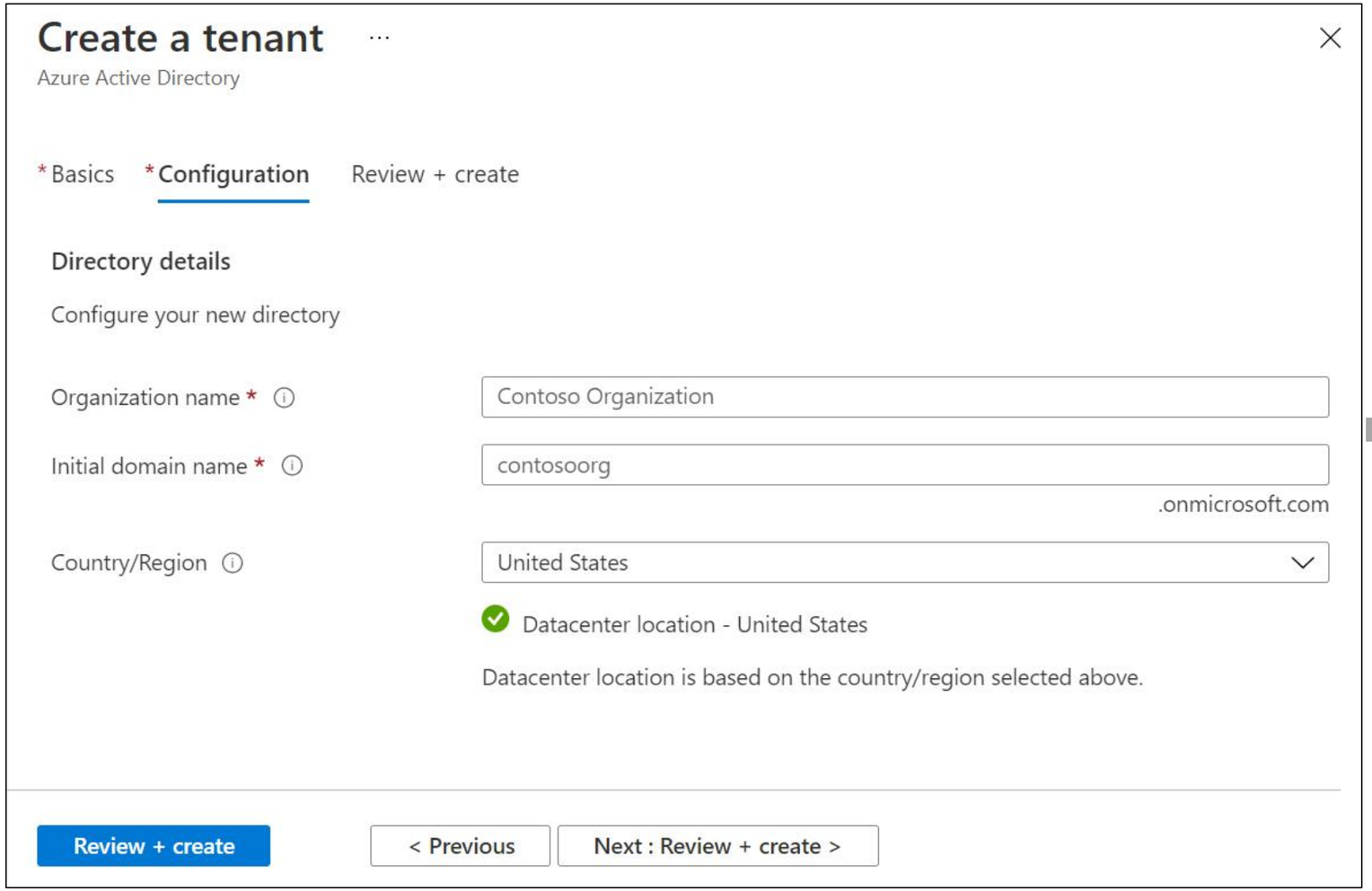
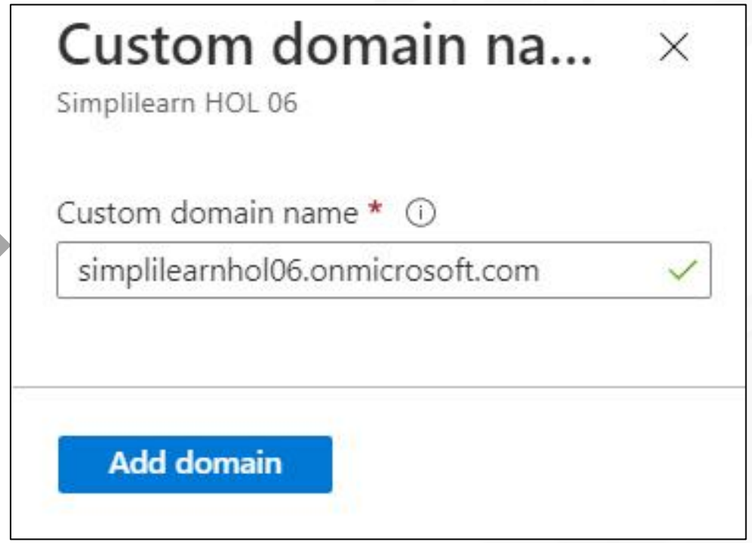
Identity Domains and Custom Domains

Azure Domain Name Service (DNS) enables to host DNS records for domains on Azure infrastructure.





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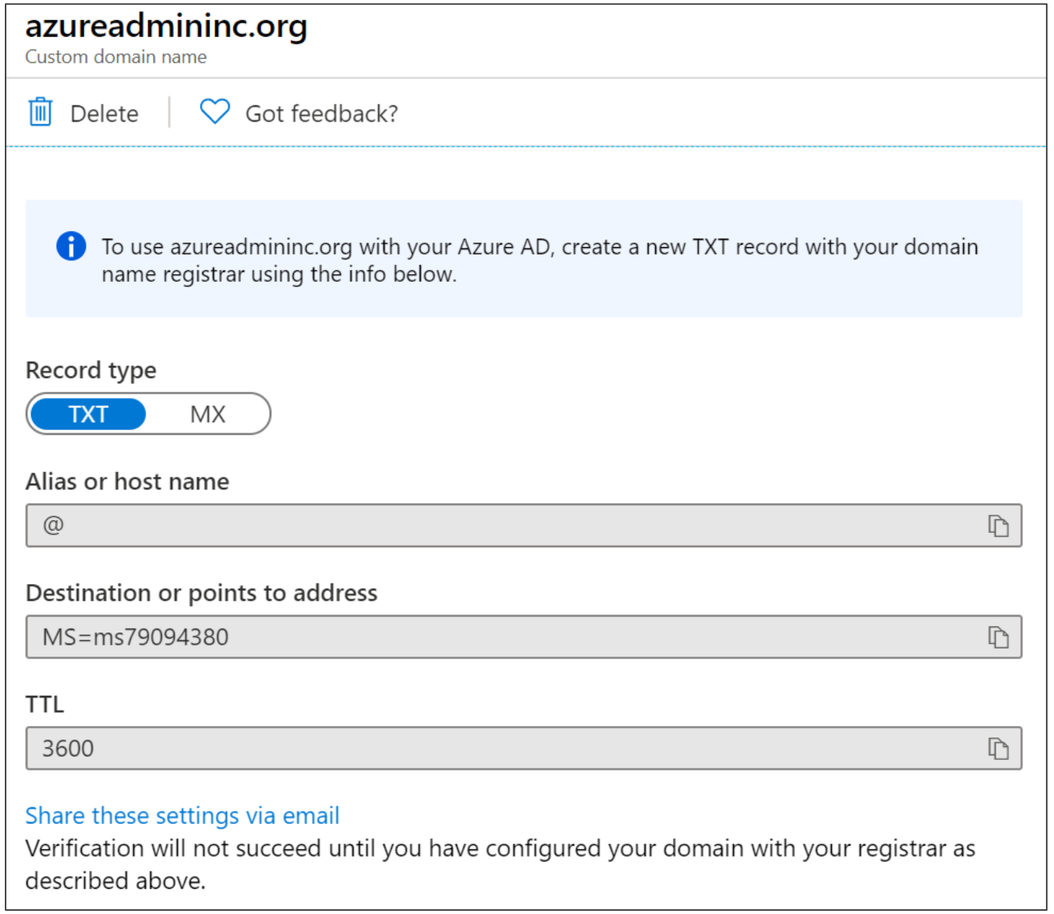
Verify Custom Domain Names

Adding a custom name to an Entra ID has the following features:



- Verification demonstrates ownership of the domain name.
- Azure's DNS record (MX or TXT) is added to the DNS zone of the company.
- Azure will query the DNS domain for the presence of the record.
- This could take several minutes or several hours.

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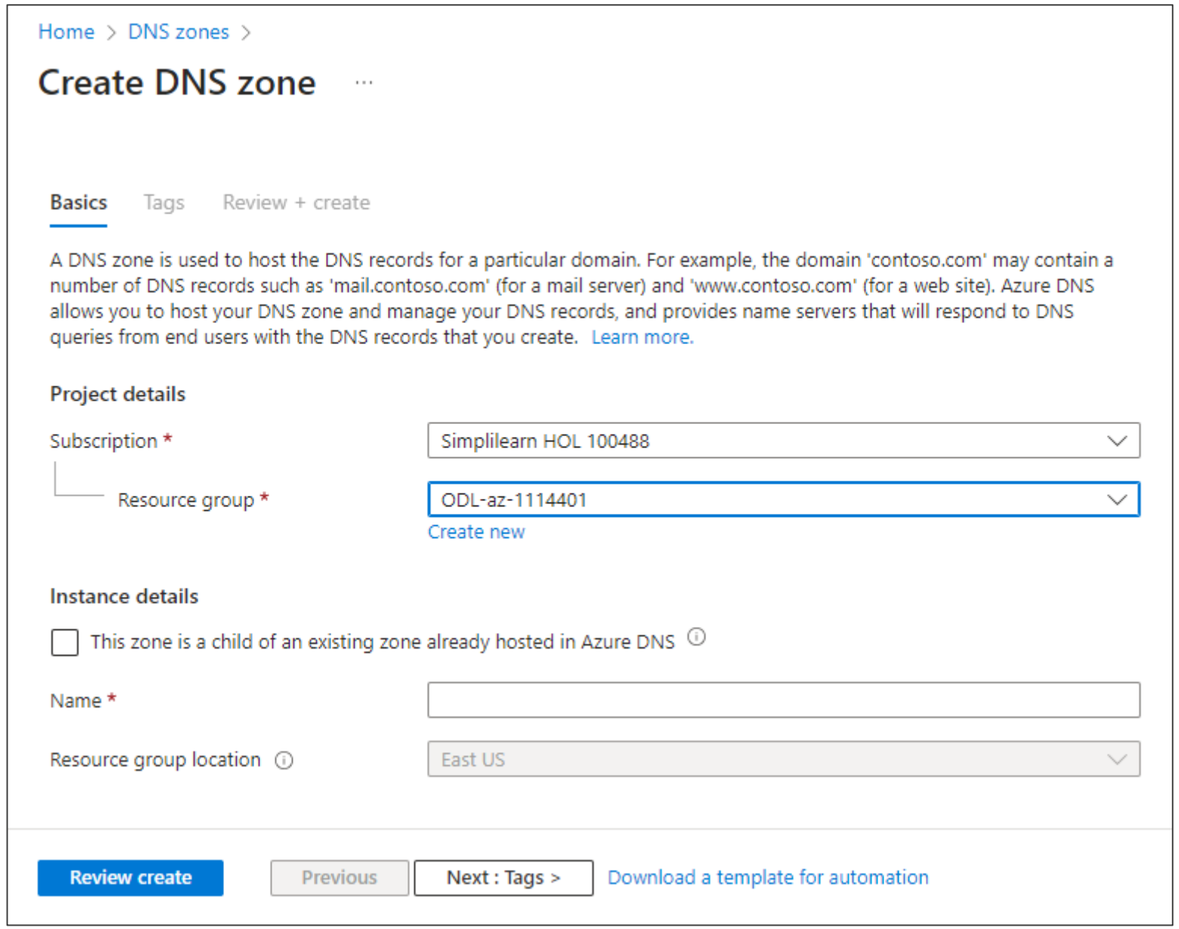
Create Azure DNS Zones

Azure DNS provides a reliable and secure DNS service to manage and resolve domain names in virtual networks.



- A DNS zone hosts the DNS records for a domain.
- Each instance is assigned a different name server address where multiple zones share the same name.
- Root or Parent domain is registered at the registrar and pointed to Azure DNS.

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Creating DNS Zones

Duration: 10 Min.

Problem Statement:

You've been tasked with creating a DNS zone and a record for user's domain.

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Assisted Practice: Guidelines

Steps to be followed:

1. Create a DNS zone
2. Add a DNS record set

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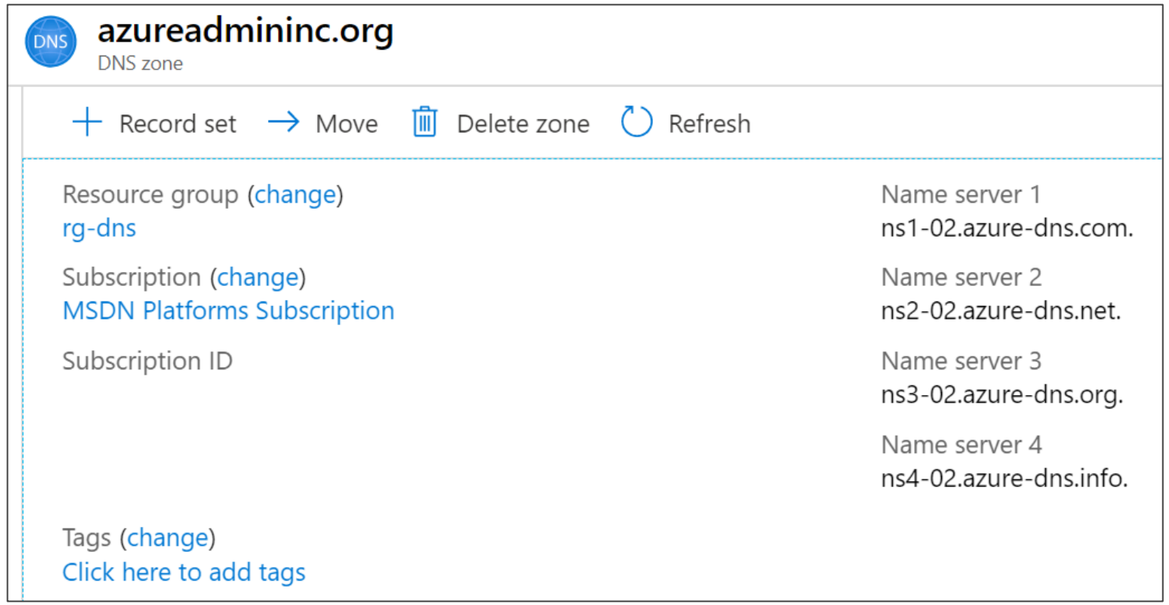
Delegate DNS Domains



To delegate the domain to Azure DNS:

- The user must use all four name server names provided by Azure DNS.
- Once the DNS zone is created, the user updates the parent registrar.
- For child zones, the user must register the DNS records in the parent domain.

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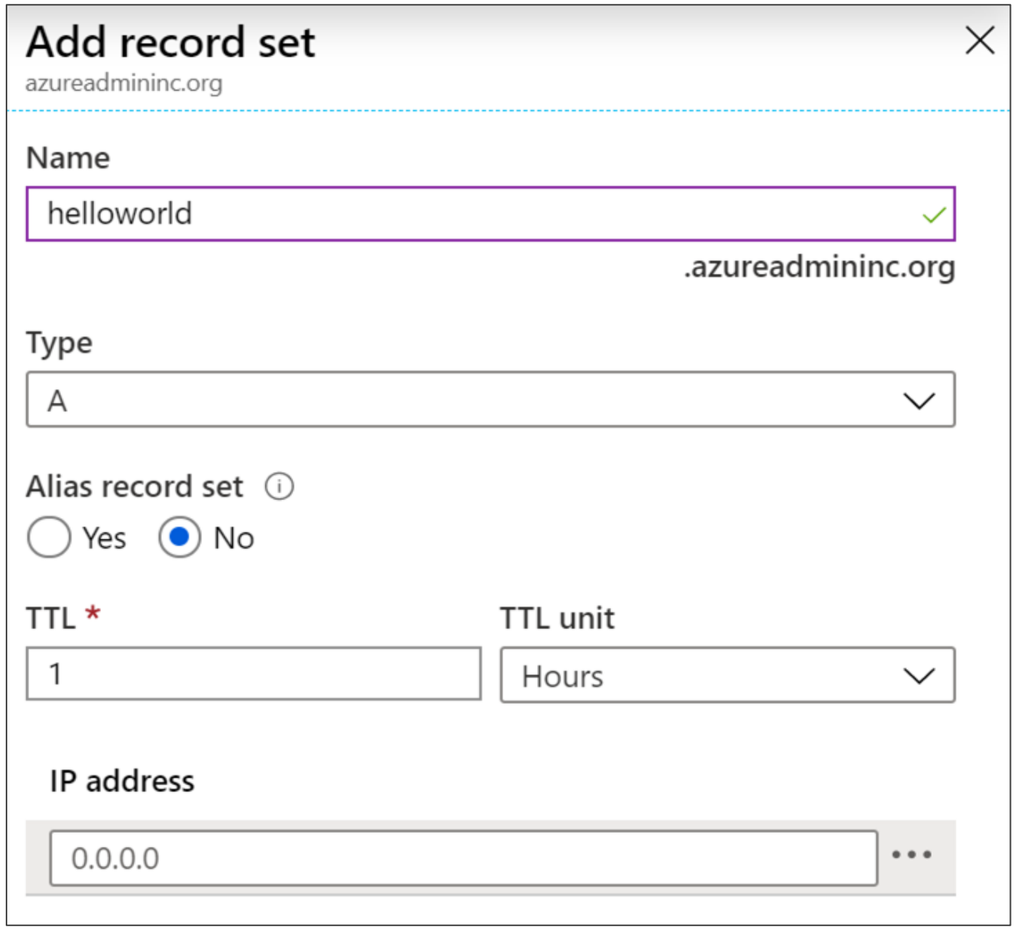
Add DNS Record Sets

It is important to grasp the difference between DNS record sets and individual DNS records.



- A record set is a collection of records in a zone that has the same name and is the same type.
- The user can add up to 20 records to any record set.
- A record set cannot contain two identical records.
- A change in the drop-down type, changes the information required.

Source: https://docs.microsoft.com/en-us/azure

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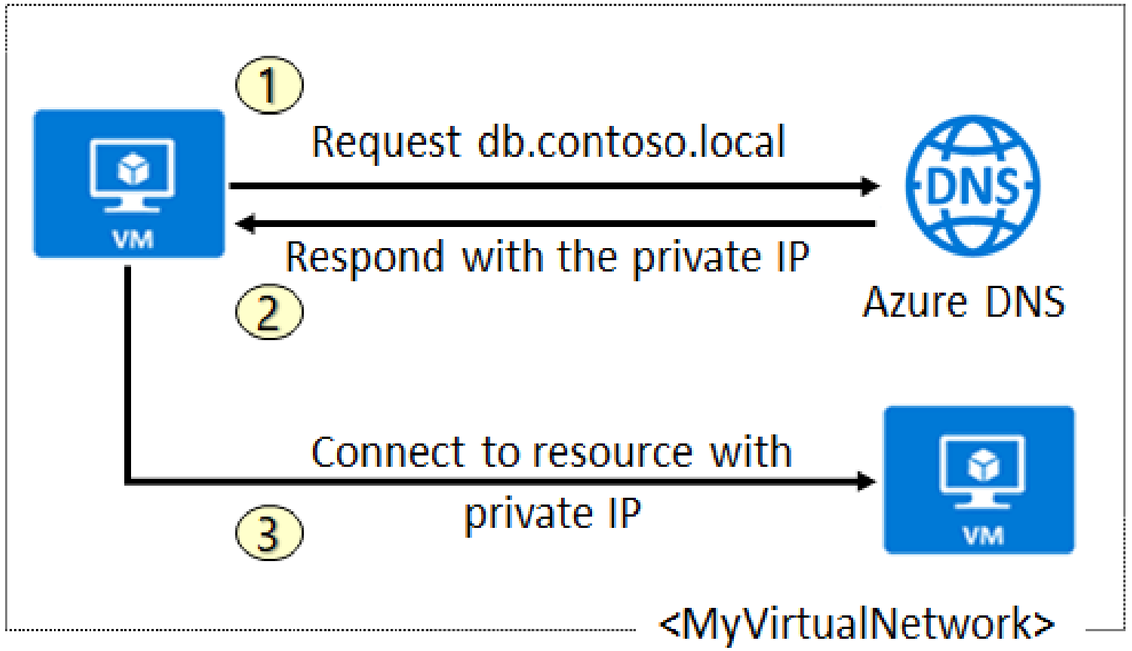
Plan for Private DNS Zones



Features of using private DNS zones:

- Provide custom domain names
- Provide name resolution for VMs within a VNet and between VNets
- Give automatic hostname record management
- Remove the need for custom DNS solutions
- Use all common DNS records types
- Available in all Azure regions

Source: https://learn.microsoft.com/en-us/answers/questions/1127021/azure-dns-zone-resolving-issue

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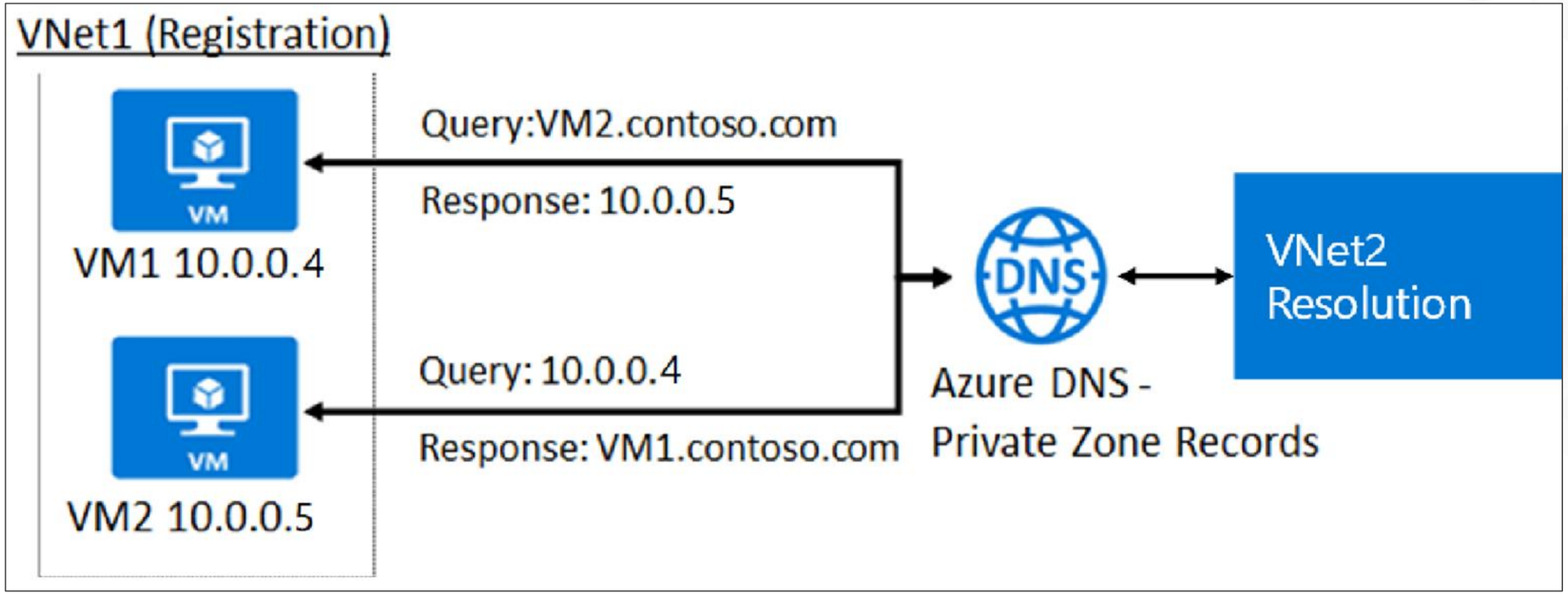
Determine Private Zone Scenarios

- DNS resolution in VNet1 is private and not accessible from the Internet.
- DNS queries across the virtual networks are resolved.
- Reverse DNS queries are scoped to the same virtual network.



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Administer Intersite Connectivity

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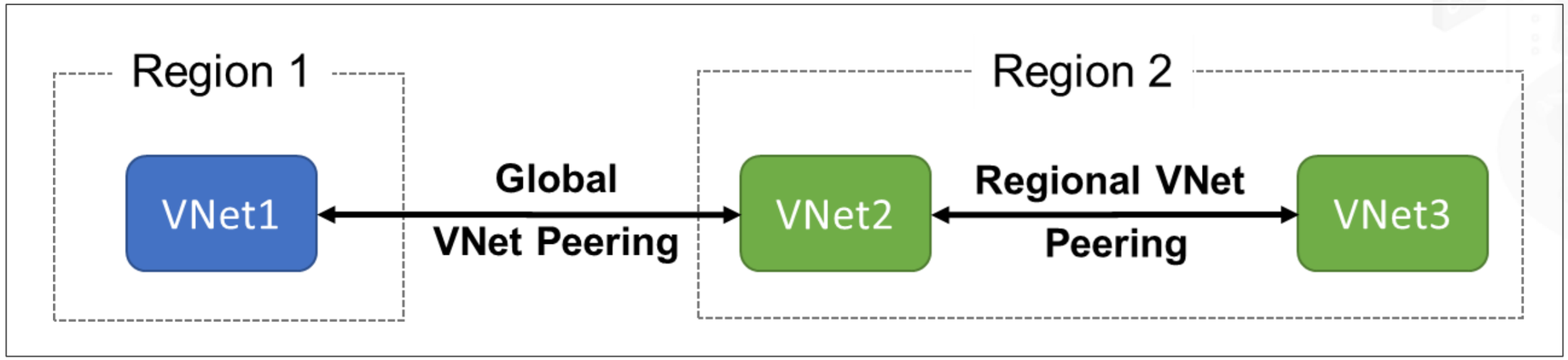
VNet Peering

Features of VNet Peering:

- VNet peering connects two Azure virtual networks.
- There are two types of peering: Regional and Global.
- Peered networks use the Azure backbone for privacy and isolation.
- The user can peer across subscriptions and tenants.
- It is easy to set up, provides seamless data transfer, and has great performance.



Source: https://docs.microsoft.com/en-us/azure/virtual-network/virtual-network-peering-overview

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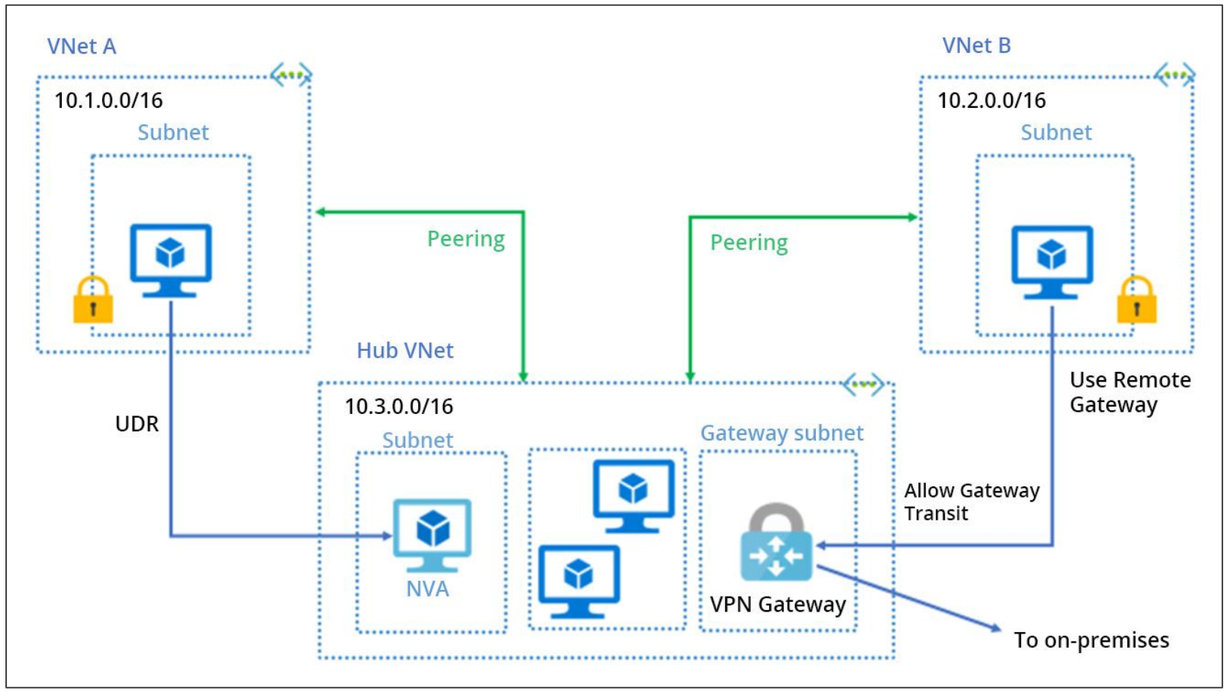
Determine Gateway Transit and Connectivity Needs

When peering virtual networks, configure a VPN gateway within the peered virtual network to serve as a transit point.



- Gateway transit allows peered virtual networks to share the gateway and get access to resources.
- No VPN gateway is required in the peered virtual network.
- Default VNet peering provides full connectivity.

Source: https://learn.microsoft.com/en-us/azure/virtual-network/virtual-network-peering-overview

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Create VNet Peering

This virtual network

Peering link name *

Traffic to remote virtual network

- (•) Allow (default)
- ( ) Block all traffic to the remote virtual network

Traffic forwarded from remote virtual network

- (•) Allow (default)
- ( ) Block traffic that originates from outside this virtual network

Virtual network gateway

- ( ) Use this virtual network's gateway
- ( ) Use the remote virtual network's gateway
- (•) None (default)

Remote virtual network

Peering link name *

Steps to configure VNet peering:

- Allow virtual network access settings
- Configure forwarded traffic settings

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Configuring VNet Peering

Duration: 30 Min.

Problem Statement:

You've been tasked with configuring VNet peering between two virtual networks.

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Assisted Practice: Guidelines

Steps to be followed:

1. Configure VNet peering on the first virtual network
2. Configure VNet peering on the second virtual network

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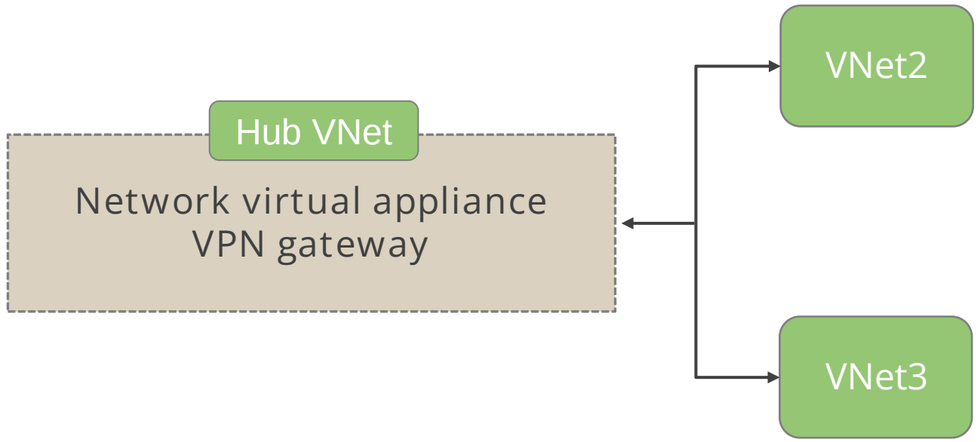
Determine Service Chaining Uses

VNet Peering is nontransitive. However, the user will configure user-defined routes and repair chaining to supply the transitivity.



- Leverage user-defined routes and service chaining to implement custom routing
- Implement a VNet hub with a network virtual appliance or a VPN gateway
- Enables direct traffic from one virtual network to a virtual appliance or virtual network gateway in a peered virtual network, through user-defined routes

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Administer Network Traffic

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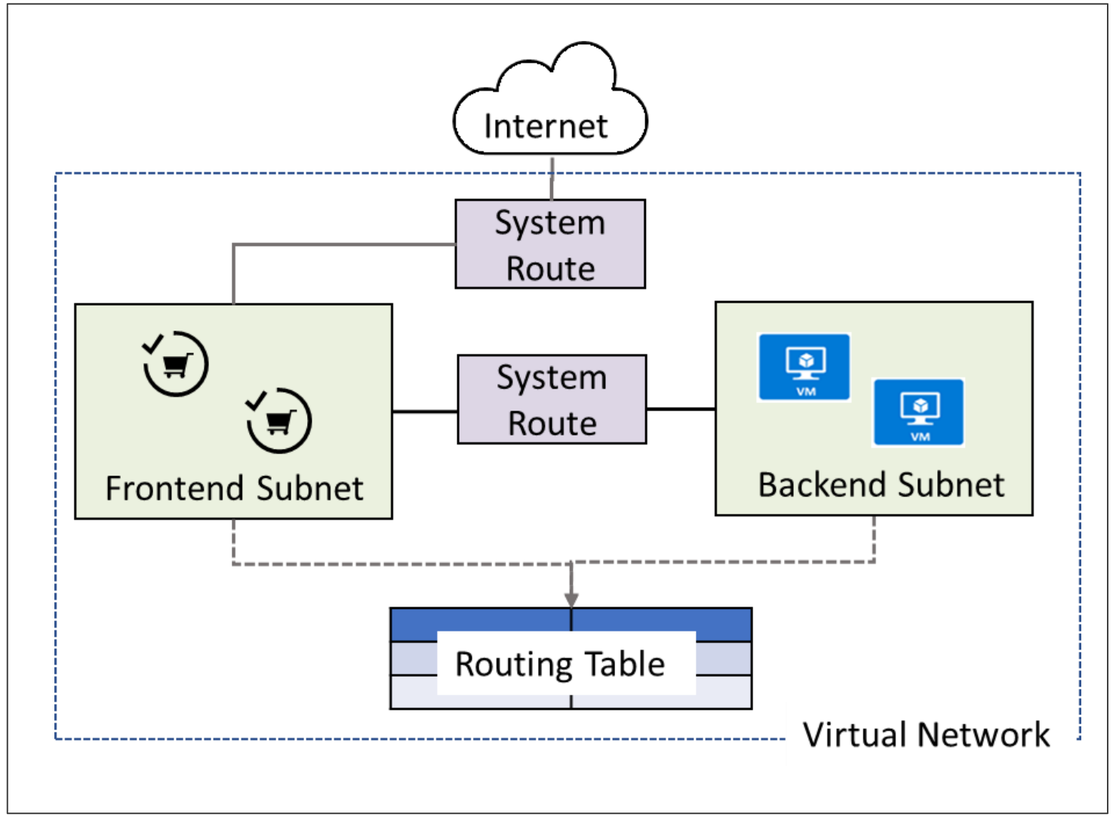
Review System Routes



System routes direct network traffic among VMs, on-premises networks, and the Internet:

- Traffic between VMs in the same subnet
- Between VMs in different subnets in the same virtual network
- Data flow from VMs to the Internet
- Communication between VMs using VNet-to-VNet VPN
- Communication between Site-to-Site and ExpressRoute through the VPN gateway

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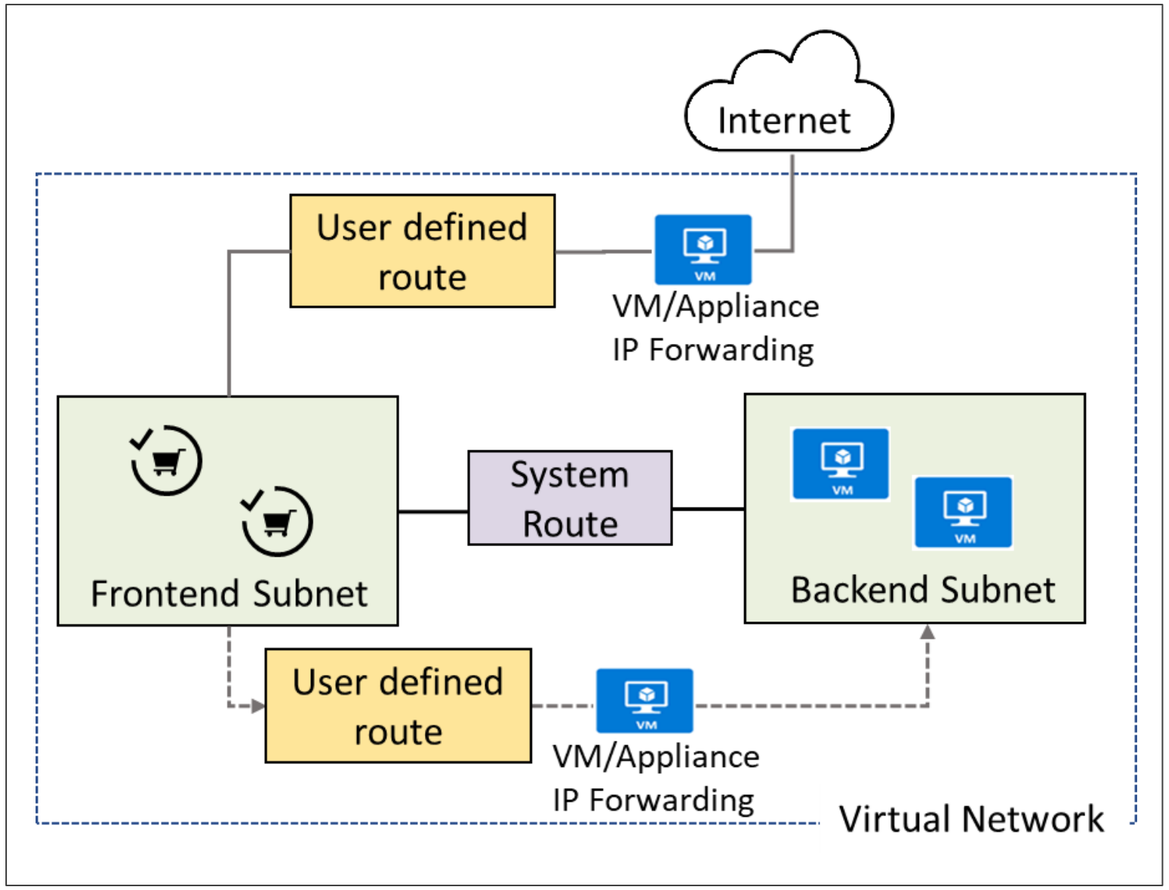
Identify User-Defined Routes



A Route table contains a set of rules, called routes, that specifies how packets should be routed in a virtual network.

User-defined routes are custom routes that control network traffic by defining routes that specify the next hop of the traffic flow.

The next hop can be a virtual network gateway, a virtual network, the internet, or a virtual appliance.

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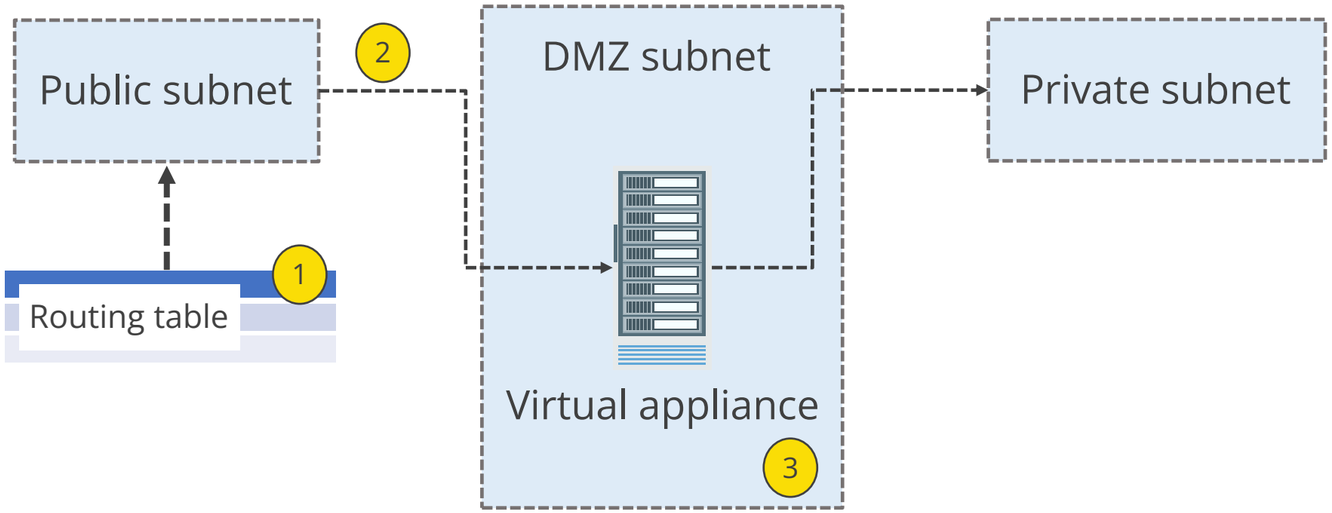
Examine a Routing Example

All traffic from the public subnet to the private subnet must go through a virtual network appliance.



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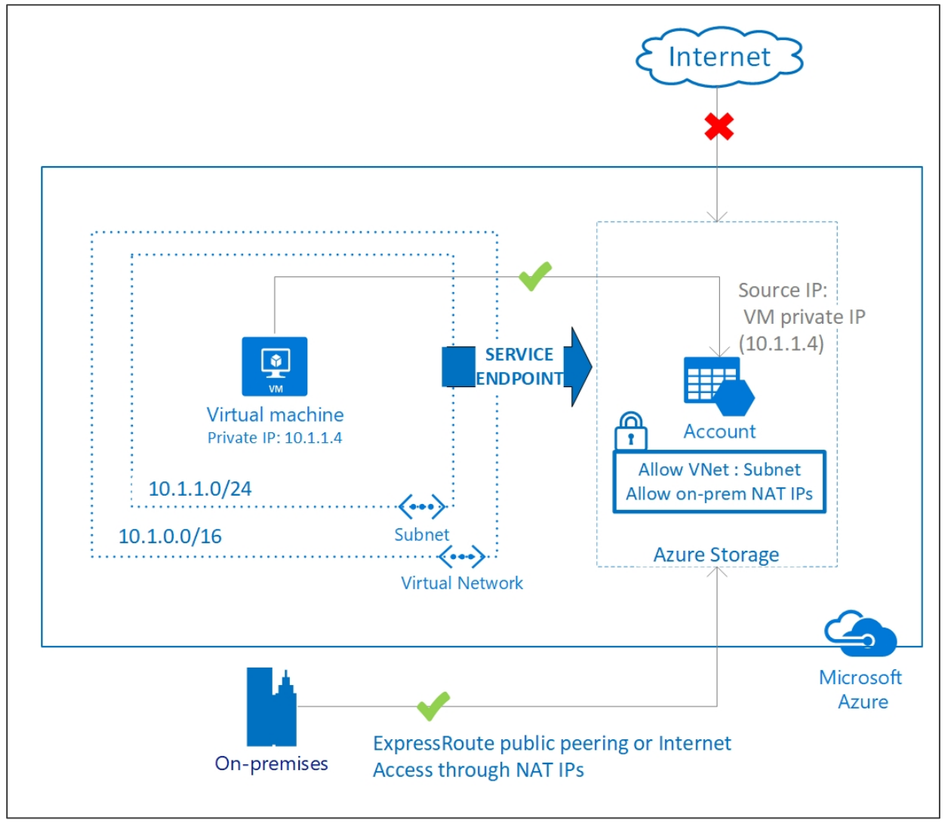
Determine Service Endpoint Uses



- Limits network access to specific subnets and IP addresses
- Improves security for the Azure service resources
- Provides optimal routing for Azure service traffic from the virtual network
- Uses the Microsoft Azure backbone network
- Provides a simple setup with less management overhead

Source: https://docs.microsoft.com/en-us/azure/virtual-network/virtual-network-service-endpoints-overview

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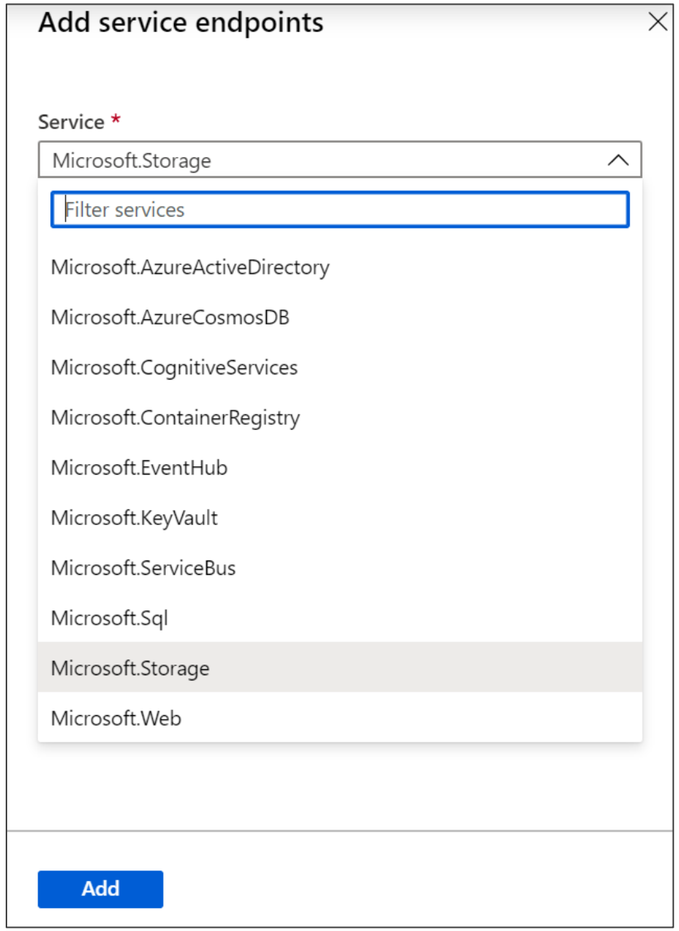
Determine Service Endpoint Services

It is easy to feature a service endpoint to the virtual network. Azure Active Directory, Azure Cosmos DB, EventHub, KeyVault, Service Bus, SQL, and Storage are some of the services available.



- There are many types of service endpoints.
- The addition of service endpoints can take up to 15 minutes to complete.

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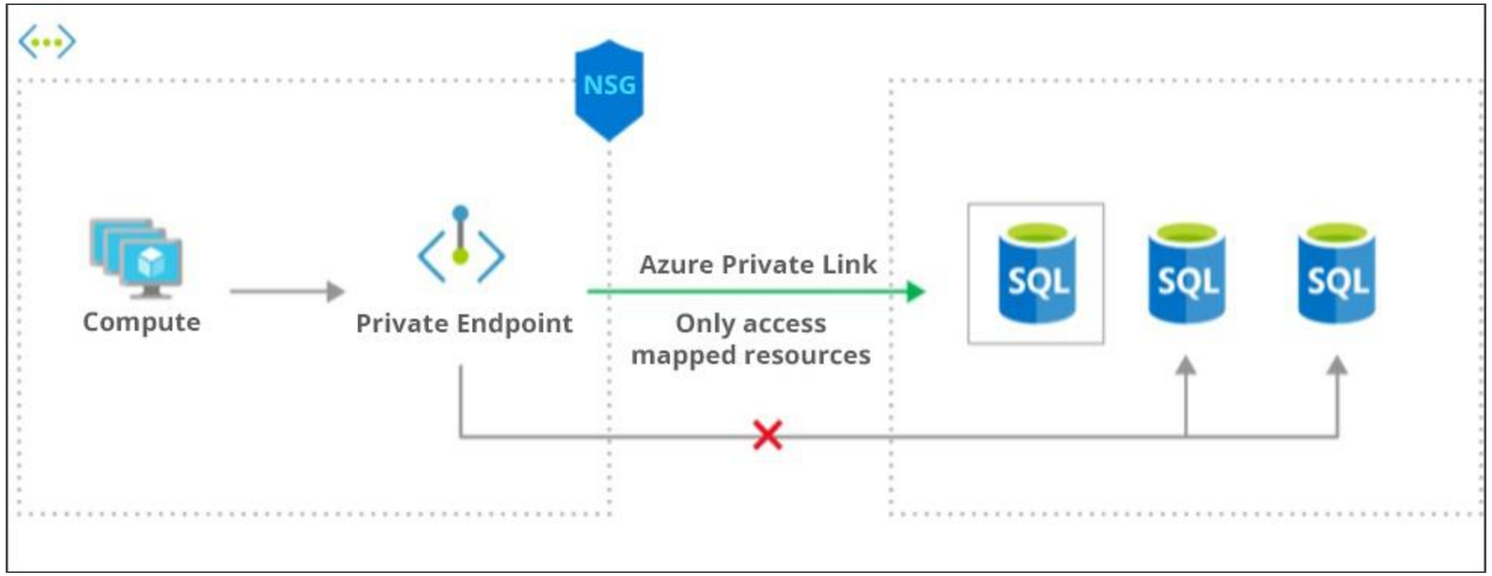
Identify Private Link Uses

- Private connectivity to services on Azure
- Traffic remains on the Microsoft network, without public internet access
- Integration with on-premises and peered networks
- The mapped resource would be accessible during a security incident within the network



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Creating a Network Virtual Appliance

Duration: 15 Min.

Problem Statement:

You've been tasked with creating a network virtual appliance to implement the user-defined route instead of system routes.

ASSISTED PRACTICE

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Assisted Practice: Guidelines

Steps to be followed:

1. Create a network virtual appliance
2. Enable IP forwarding for the network interface
3. Enable IP forwarding for the appliance
4. Create a virtual machine

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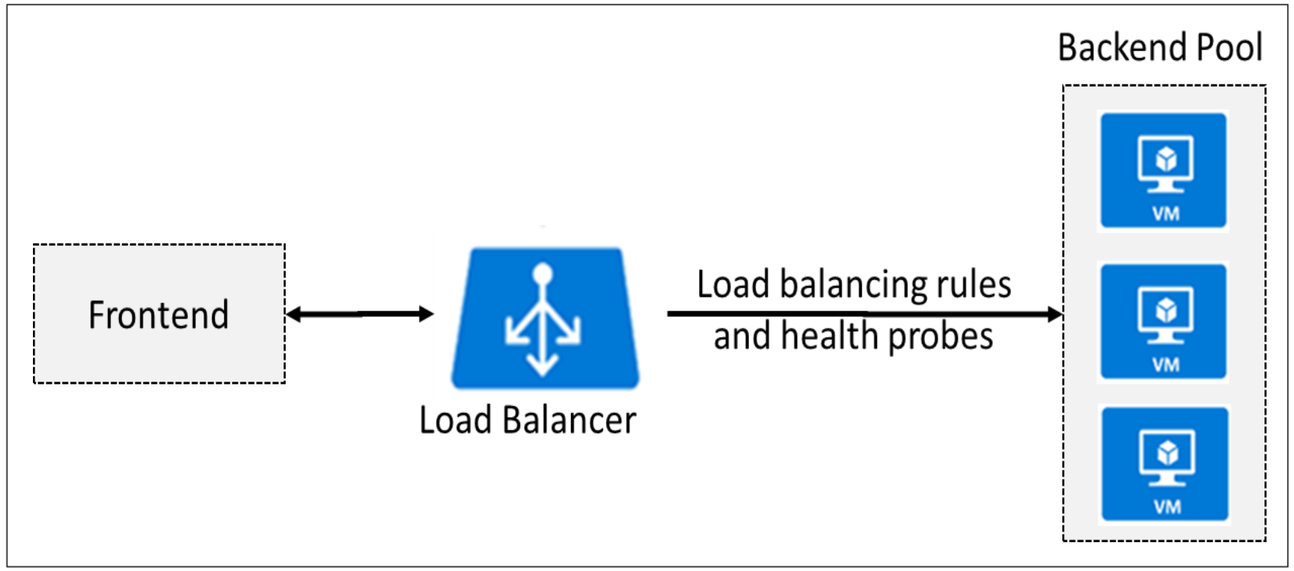
Determine Azure Load Balancer Uses

Azure Load Balancer delivers high availability and network performance to applications. The load balancer distributes inbound traffic to backend resources using load balancing rules and health probes.



- Distributes inbound traffic to backend resources using load balancing rules and health probes
- Can be used for both inbound or outbound scenarios
- Has two types: Public and Internal

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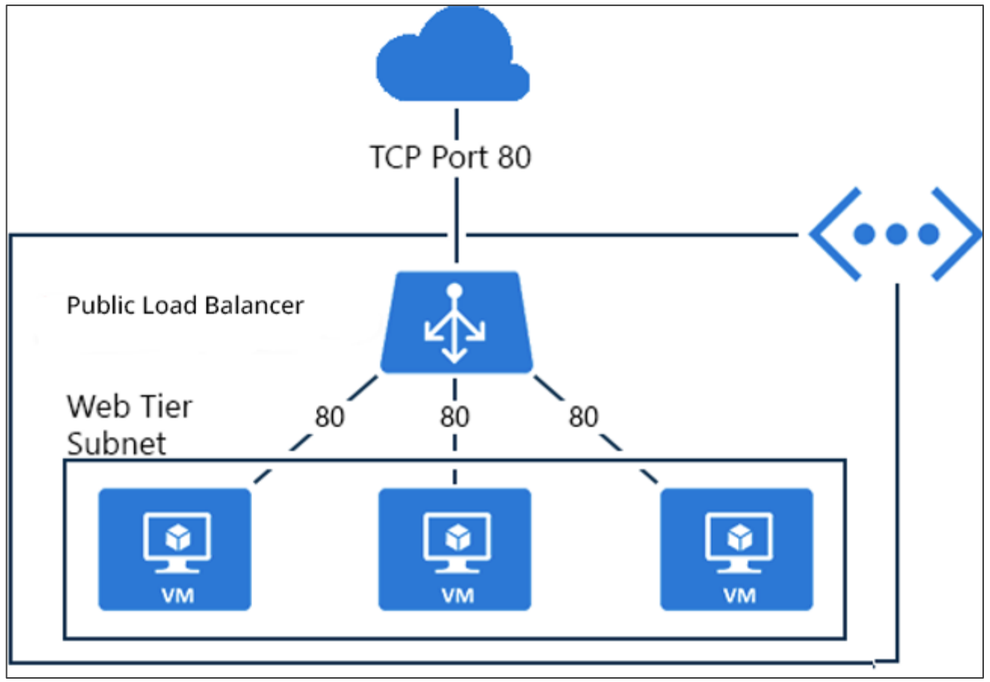
Implement a Public Load Balancer

It helps to:

- Map public IP addresses and port number of incoming traffic to the VM's private IP address and port number and vice versa
- Apply load balancing rules to distribute traffic across VMs or services



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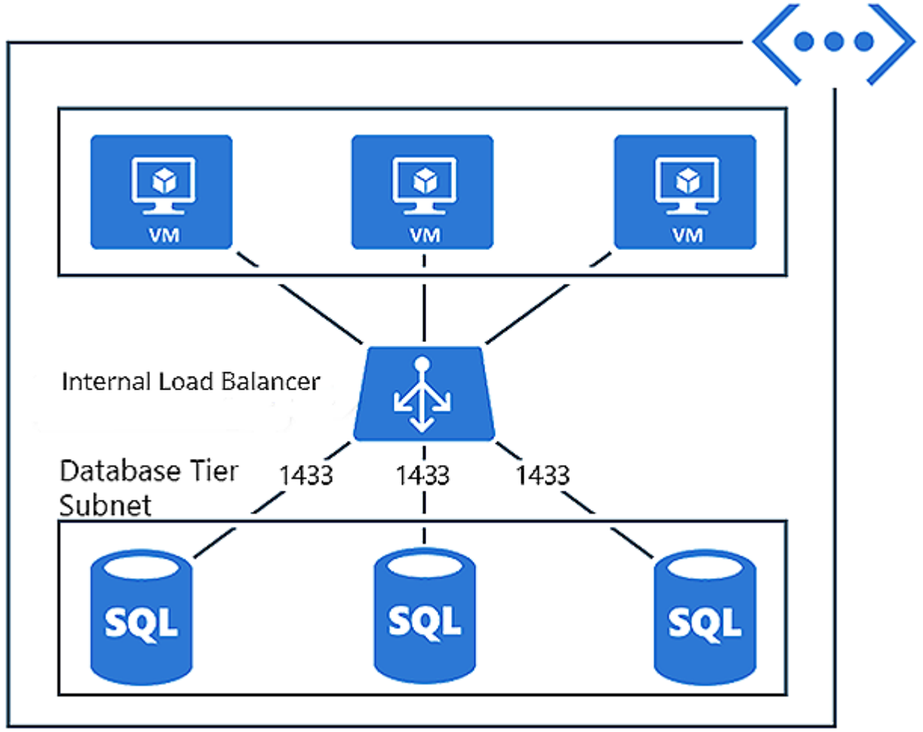
Implement an Internal Load Balancer

The advantages of implementing an internal load balancer are:



- Directs traffic only to resources inside a virtual network or that use a VPN to access the Azure infrastructure
- Does not directly expose front-end IP addresses and virtual networks to an Internet endpoint
- Enables load balancing within a virtual network for cross-premises virtual networks, multi-tier applications, and line-of-business applications

Source: https://docs.microsoft.com/en-us/azure/load-balancer/load-balancer-overview

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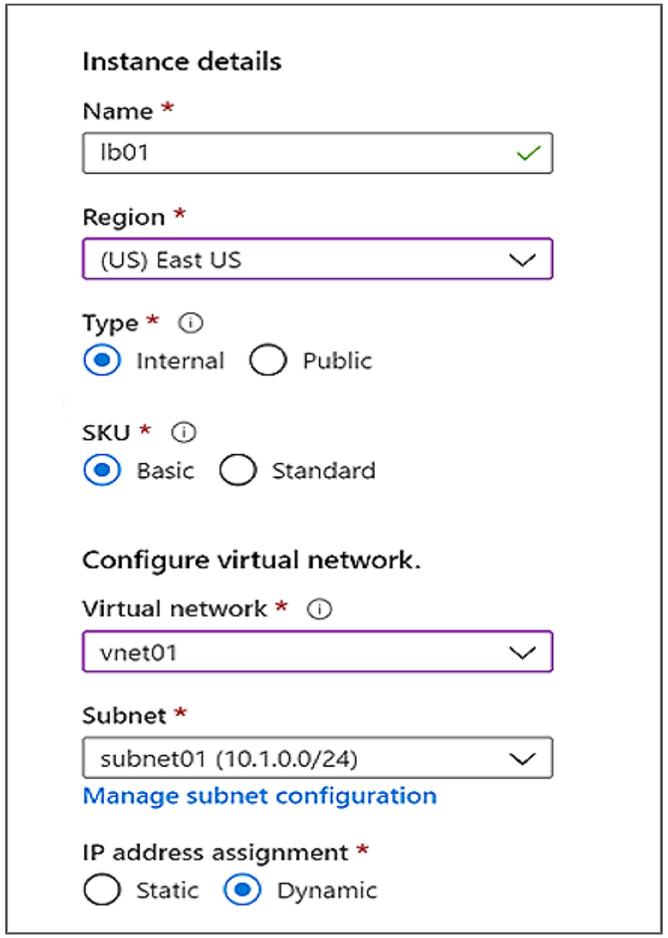
Determine Load Balancer SKUs

The user must choose the Load Balancer SKU:



Feature	Basic SKU	Standard SKU
Back-end pool	Up to 300 instances	Up to 5000 instances
Health probes	TCP and HTTP	TCP, HTTP, and HTTPS
Availability zones	Not available	Zone-redundant and zonal frontends for inbound and outbound traffic
Multiple frontends	Inbound only	Inbound and outbound
Secure by default	Open by default. NSG optional	Closed to inbound flows unless allowed by NSG; Internal traffic from the virtual network to the internal load balancer is allowed
SLA	Not available	99.99%

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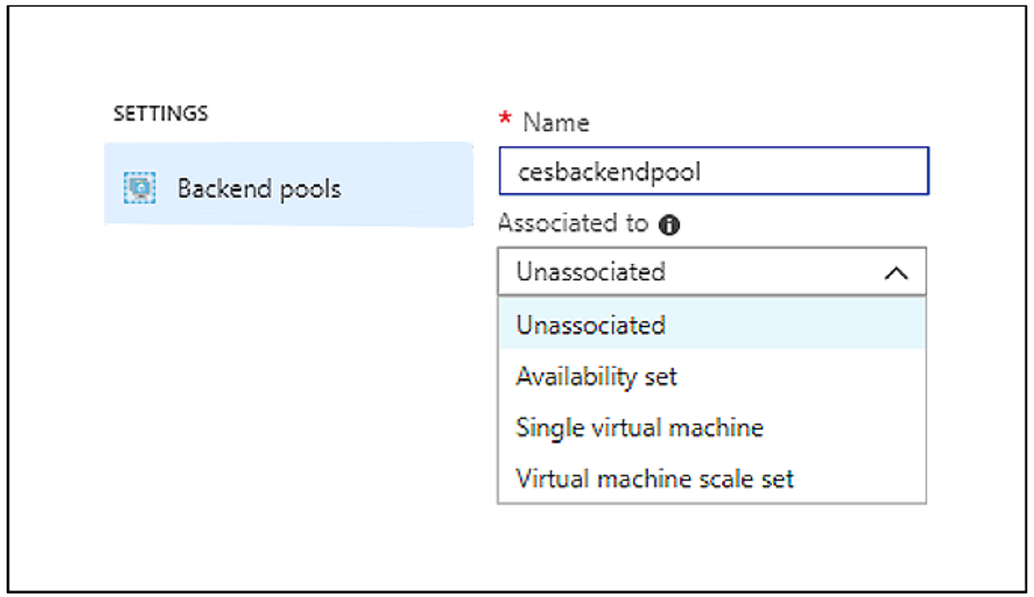
Create Backend Pools

To distribute traffic, a backend address pool contains the IP addresses of virtual NICs that are connected to the load balancer.



SKU	Backend pool endpoints
Basic SKU	VMs in a single availability set or VM scale set
Standard SKU	Any VM in a single virtual network, including a blend of VMs, availability sets, and VM scale sets

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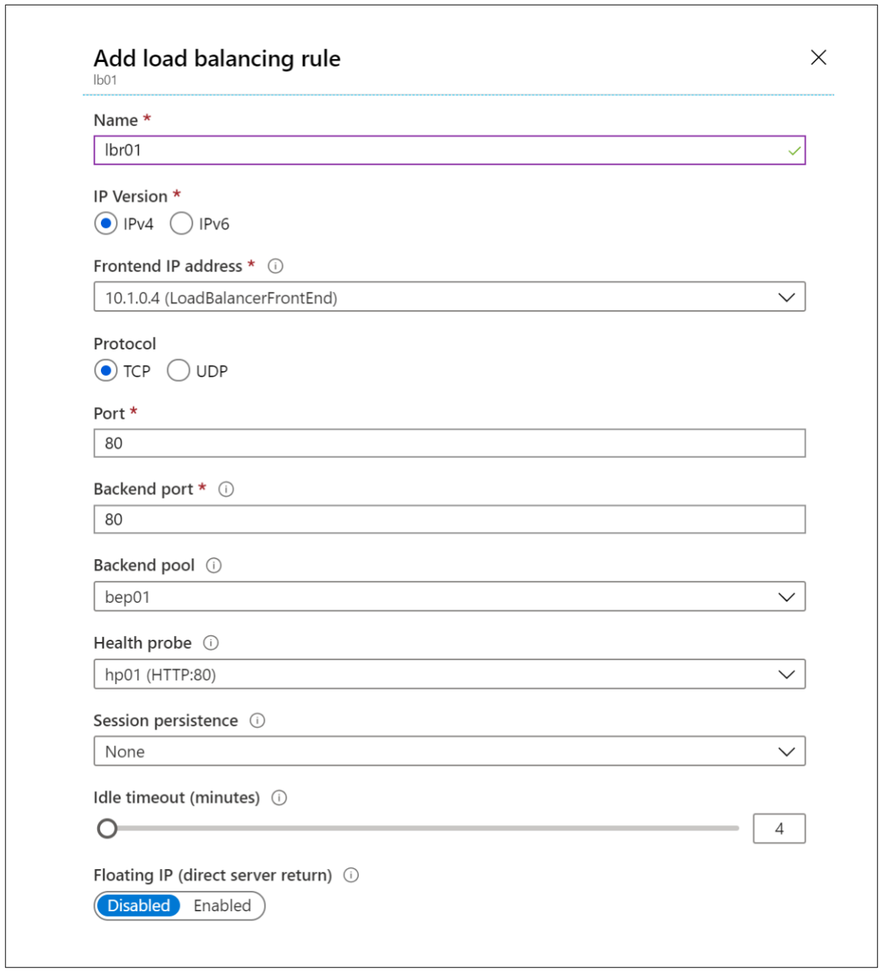
Create Load Balancer Rules

A load balancer rule defines how traffic is distributed to the backend pool. Its features include:



- A frontend IP and the port combination can be mapped to a set of backend pools and port combinations.
- Rules can be combined with NAT rules.
- A NAT rule is explicitly attached to a VM (or network interface) to complete the path to the target.

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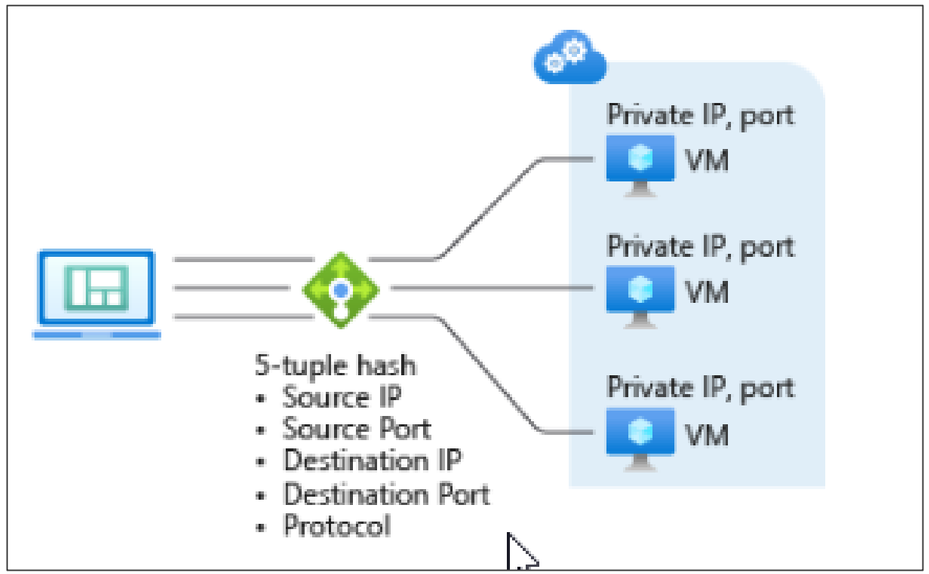
Configure Session Persistence

Azure Load Balancer distributes network traffic equally among multiple VM instances. It uses a five-tuple hash to map traffic to available servers, and it provides stickiness only within a transport session.



- Session persistence specifies how client traffic is handled.
- None (default) requests can be handled by any virtual machine.
- Client IP requests will be handled by the same virtual machine.
- Client IP and protocol specify that successive requests from the same address and protocol will be handled by the same virtual machine.

Source: https://learn.microsoft.com/en-us/azure/load-balancer/distribution-mode-concepts

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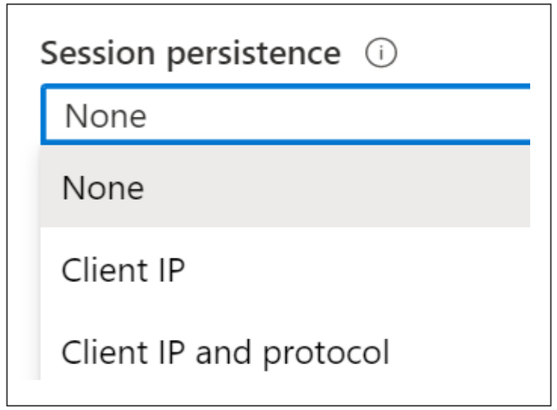
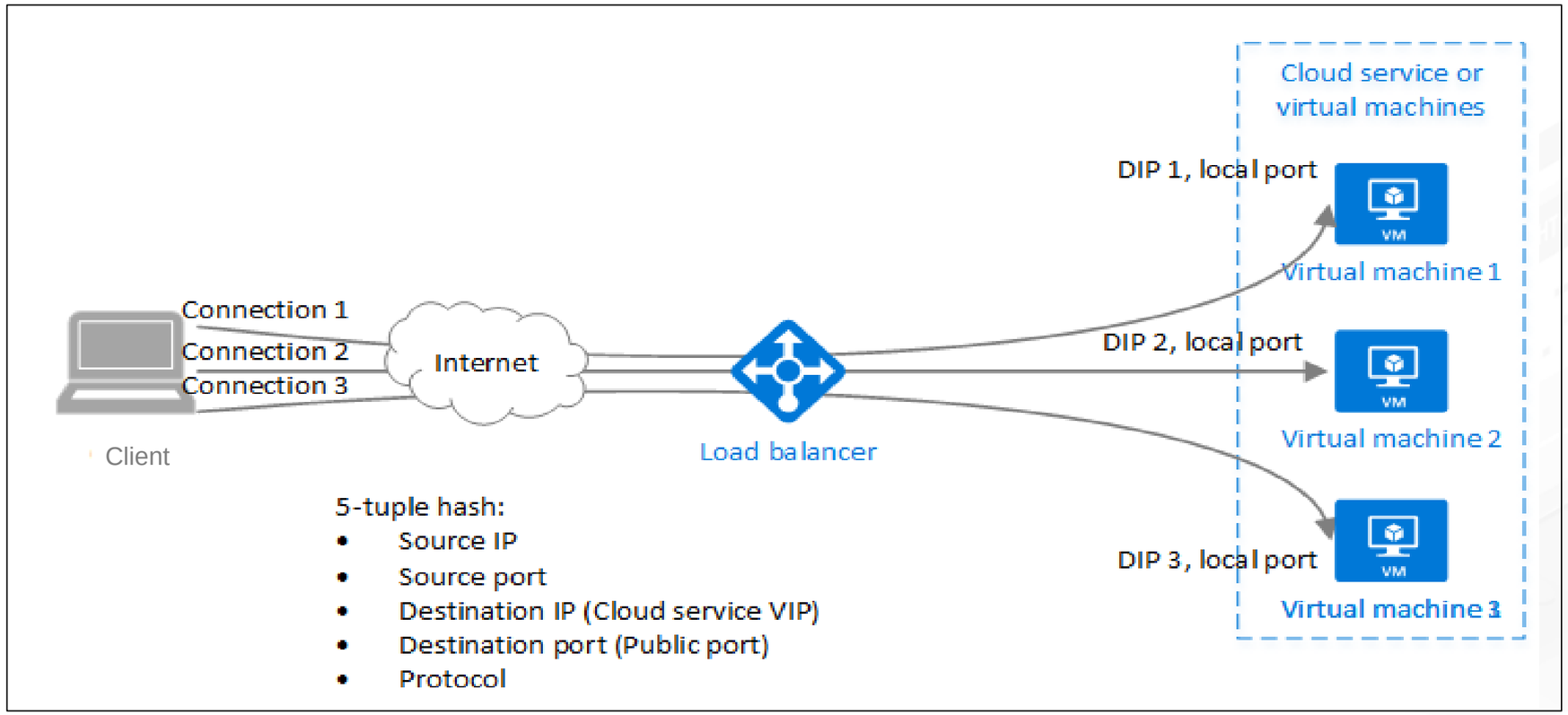
Configure Session Persistence

The five-tuple hash for mapping traffic to available servers is illustrated below:





Source: https://docs.microsoft.com/en-us/azure

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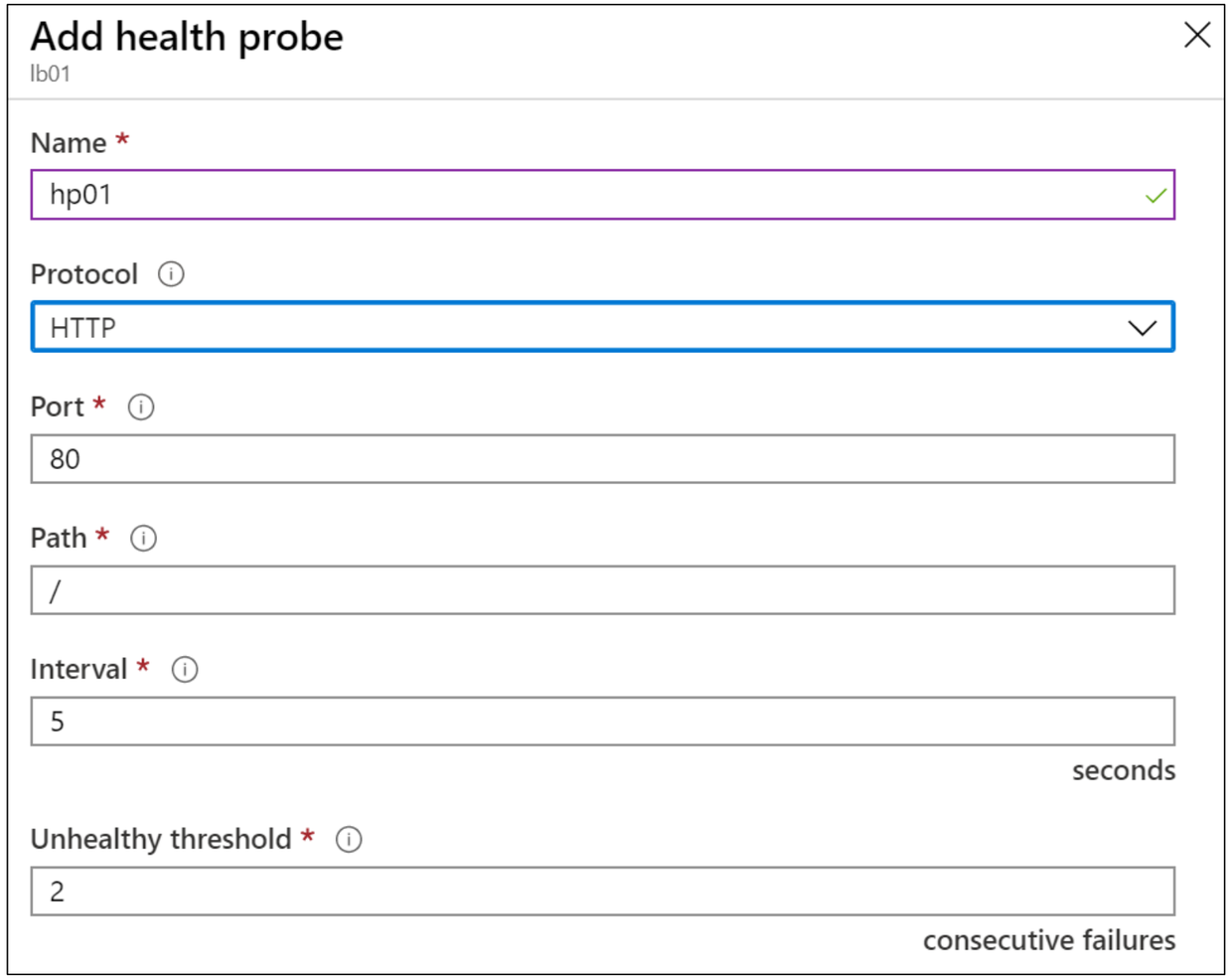
Create Health Probes



Features:

- The load balancer can monitor the status of an app.
- The VMs can be added or removed from the load balancer rotation based on their response to health checks.
- An HTTP custom probe (preferred) pings every 15 seconds.
- TCP custom probe tries to establish a successful TCP session.

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Configuring a Public Load Balancer

Duration: 30 Min.

Problem Statement:

You’ve been tasked with configuring a public load balancer.

UNASSISTED PRACTICE

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Assisted Practice: Guidelines

Steps to be followed:

1. Create a virtual network
2. Create a NAT gateways
3. Create a load balancer
4. Create virtual machines
5. Install IIS
6. Test the load balancer

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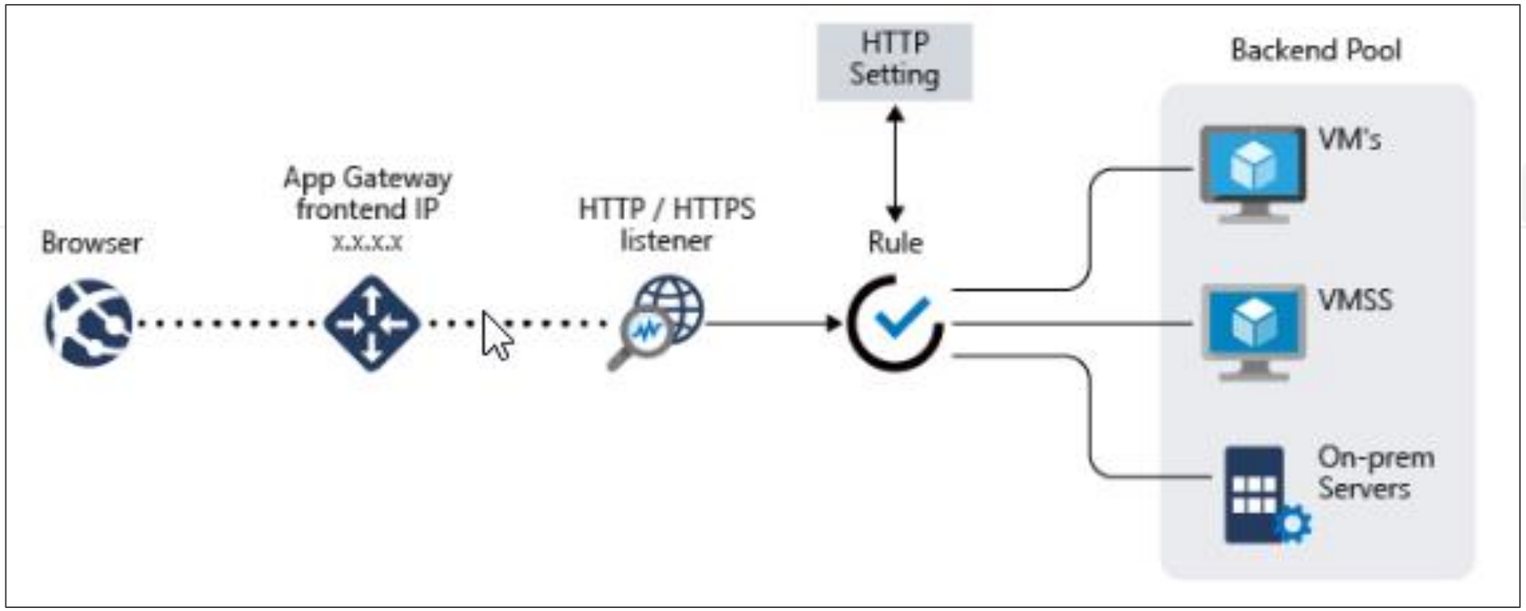
Implement Application Gateway

Features of the Application Gateway:

- Manages web app requests
- Routes traffic to a pool of web servers based on the URL of a request
- Includes web servers that can be Azure virtual machines, Azure virtual machine scale sets, Azure App Service, and even on-premises servers



Source: https://learn.microsoft.com/en-us/azure/application-gateway/features

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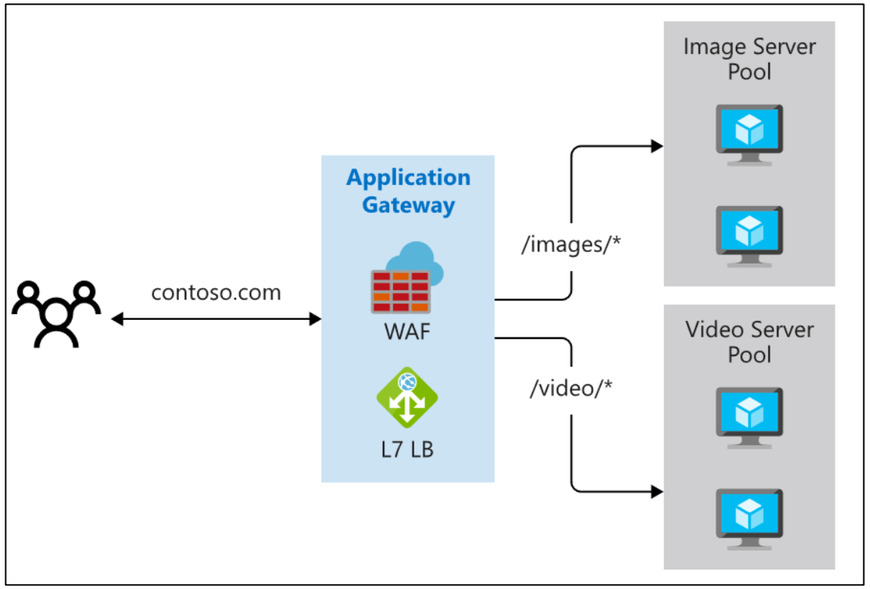
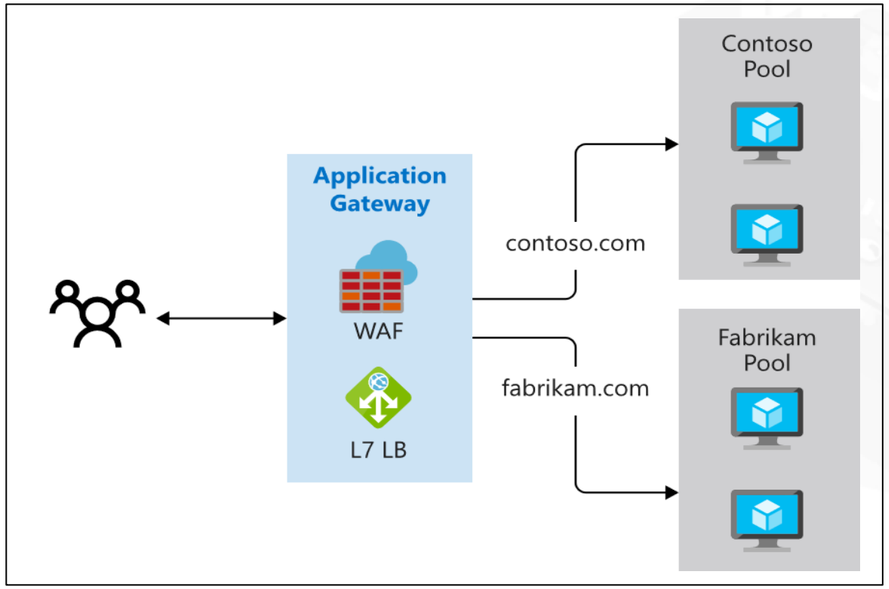
Determine Application Gateway Routing

Clients send requests to web apps via the gateway's IP address or DNS name. The gateway uses rules to route these requests to a specific server in the backend pool, with two main traffic routing methods.

Path-based routing



Multiple-site routing



Source: https://learn.microsoft.com/en-us/training/modules/load-balance-web-traffic-with-application-gateway/2-routing-traffic-with-application-gateway

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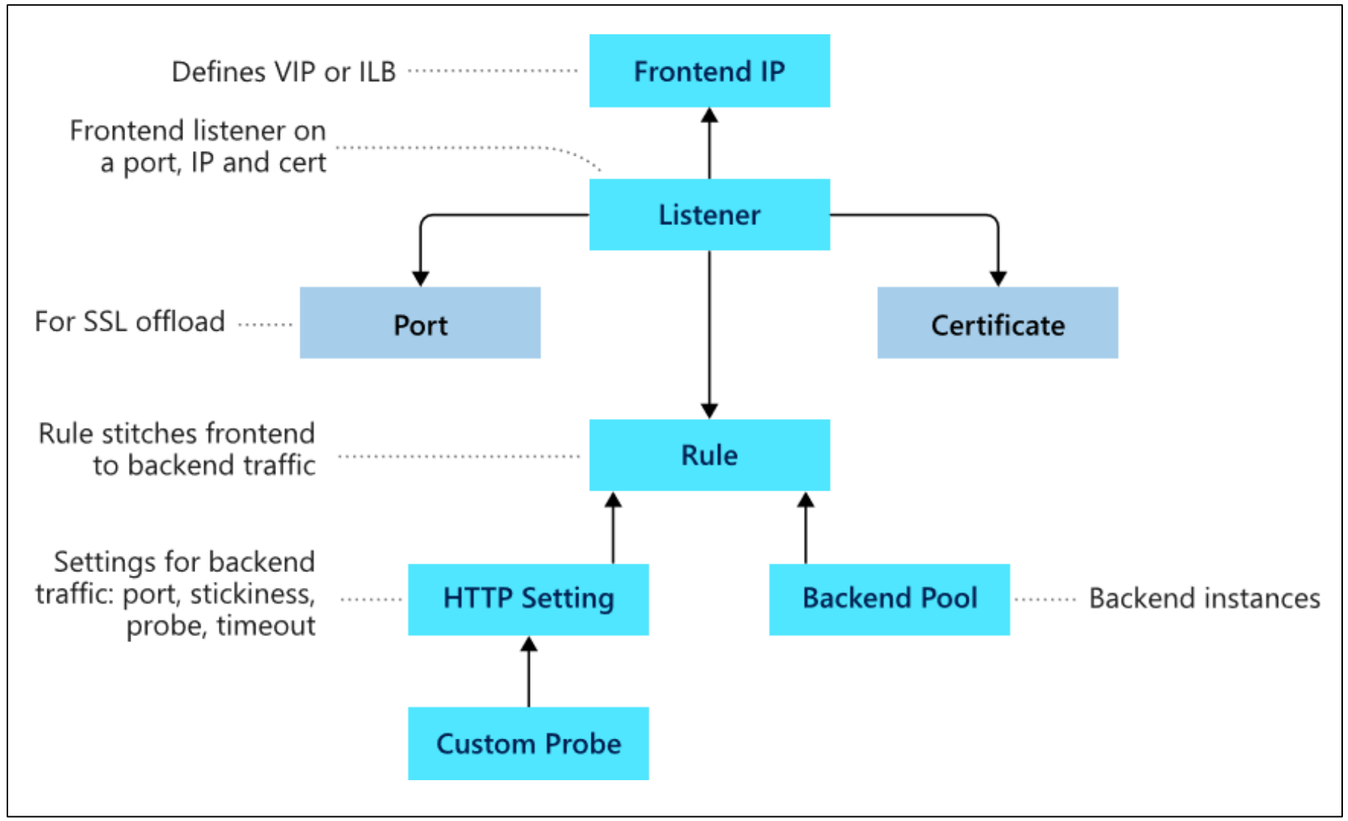
Set Up Application Gateway Components

Application Gateway has a series of components that combine to route requests to a pool of web servers and check the health of these web servers.



- Frontend IP
- Listeners
- Routing rules
- Backend pools
- Web application firewall (optional)
- Health probes

Source: https://learn.microsoft.com/en-us/training/modules/load-balance-web-traffic-with-application-gateway/4-create-configure-application-gateway

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Troubleshoot Network Connectivity

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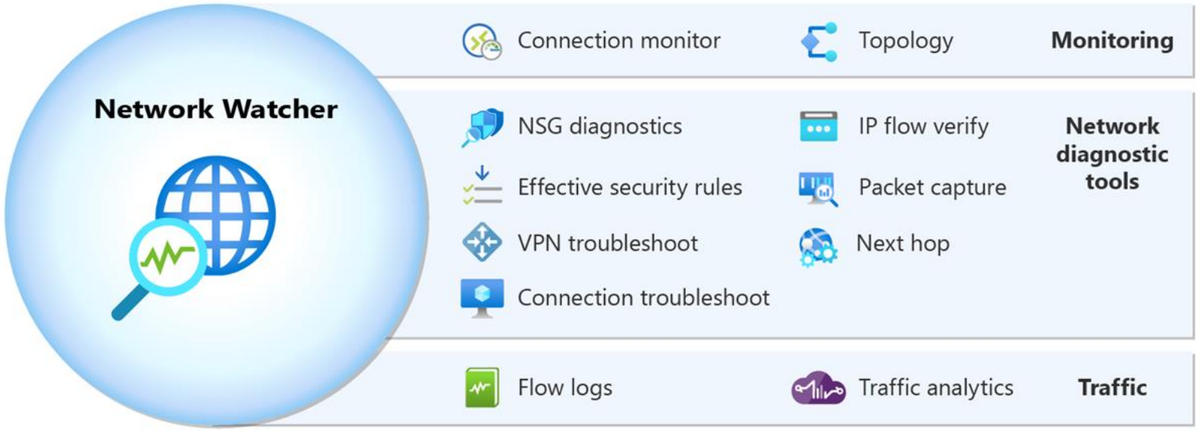
Network Watcher

It offers a set of tools to observe, analyze, and view metrics and toggle logs for Azure IaaS (Infrastructure-as-a-Service) assets.



It consists of three major sets of tools and capabilities:

- Monitoring
- Network diagnostic tools
- Traffic

Source: https://learn.microsoft.com/en-us/azure/network-watcher/network-watcher-overview

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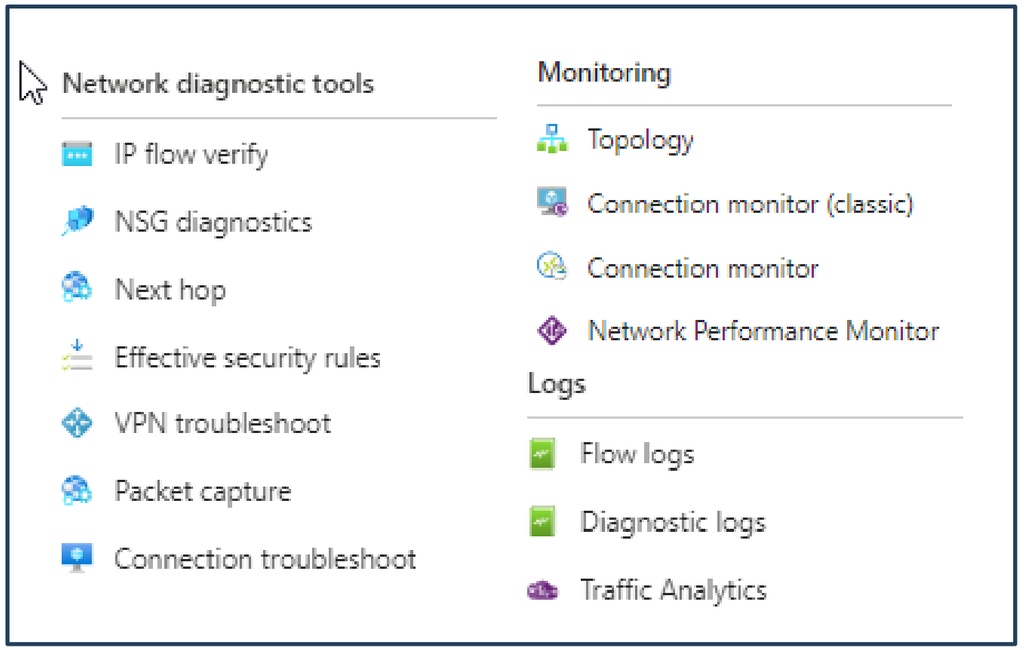
Network Watcher Features



- IP flow verify diagnoses connectivity issues
- Next hop determines if traffic is being correctly routed
- VPN troubleshoot is used to diagnose gateways and connection
- NSG flow logs map IP traffic through a network security group
- Connection troubleshoot shows connectivity between source VM and destination
- Topology generates a visual diagram of resources

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Key Takeaways

- An Azure Virtual Network (VNet) is a representation of a network in the cloud. It is a logical isolation of the Azure cloud dedicated to the subscription.
- A network security group contains a list of security rules that allow or deny inbound or outbound network traffic.
- Azure DNS enables users to host DNS records for their domains on the Azure infrastructure.

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Implementing Path-Based Routing with Application Gateway

Duration: 15 Min.

Project agenda: To implement path-based routing using the Azure Application gateway

Description: To create routing based on the path provided in the URL to perform path-based routing

Perform the following:

Create Application gateway, backend pool, and front-end IP. Then, configure the routing rules based on the path given in the URL.

LESSON-END PROJECT

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